

System Design Specification

for

Employee Performance Review (EPR) Dashboard and Workflow System

Version 7.0

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23 April 2026

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Document Control

Version	Date	Author	Summary
1.0	10/16/2025	Momoka Aung, Chiemela Eziechile	Initial draft.
2.0	10/20/2025	Hoang Le (Dempsey)	Updated 3.2.3 (HR/EPR pages); added KPI definitions and why they matter; UX notes; added flow caption.
3.0	12/01/2025	Momoka Aung	Added detailed information of System Strategy and Architecture part.
4.0	12/03/2025	Momoka Aung	Updated the documents with work flow diagrams and automation flow diagrams.
5.0	12/03/2025	Momoka Aung	Update Box Upload function to the current version.
6.0	03/23/2026	Momoka Aung	Added Section 9 Database Design.
7.0	04/20/2026	Hoang Le (Dempsey)	Updated Section 9 User Interface with expanded Power BI dashboard documentation for Personnel Matters and Vacancies & Recruitment, including Smartsheet data, intake forms, tracker screenshots, and dashboard page descriptions.
8.0	04/28/2026	James San	Added setup for all projects python scripts and AWS build and deploy steps.

1. Introduction

1.1. Purpose

The purpose of this document is to define the technical architecture and detailed design of the Intelligent HR Analytics Platform developed for the Santa Barbara County Public Defender's Office (SBCPDO). This Software Design Specification (SDS) translates the approved Software Requirements Specification (SRS) into a concrete system design that guides system implementation, integration, testing, and maintenance.

1.2. Scope of Coverage

This SDS describes the design of the platform's core components, including workflow automation for Employee Performance Reviews (EPR), Personnel Matters, Vacancies and Recruitments, and Employee Separations. It also outlines document management integration with Box, reporting through Power BI, and role-based access controls. Business policies and operational procedures are defined in the SRS and are not repeated here.

1.3. Document Conventions

This document follows a hierarchical numbering structure (e.g., 1.0, 1.1, 1.2). Formatting conventions are used to distinguish elements of the design:

- **Bold text** indicates section titles or key architectural components.
- *Italic text* refers to external documents such as the SRS.
- `Monospaced text` represents field names, configuration values, or identifiers.

1.4. Intended Audience

This document is intended for developers, system architects, testers, and technical staff responsible for implementing or maintaining the system. Project managers and administrators may also reference this document to understand the system's architecture. Non-technical stakeholders should refer to the SRS for functional and business-level descriptions.

1.5. System Overview

The Intelligent HR Analytics Platform is a web-based system that automates and centralizes key HR workflows within SBCPDO. The system supports performance reviews, personnel case management, recruitment tracking, and employee separations through integrated workflows and reporting dashboards.

The architecture consists of several logical layers: a workflow layer for HR processes, an integration layer connecting external services, a data layer for structured records and document links, an analytics layer for reporting, and a security layer enforcing role-based access control. Together, these components provide a scalable and maintainable platform for managing HR operations and generating leadership insights.

2. Design Considerations

This section describes the key assumptions, constraints, and design goals that influence the architecture and implementation of the Intelligent HR Analytics Platform.

2.1. Assumptions and Dependencies

Technologies and platforms used in the system include:

- **Smartsheet:** Serves as the primary workflow engine for tracking EPRs, personnel matters, vacancies and recruitments, and employee separations.
- **Box:** Used as the secure document repository for finalized HR documents and records.
- **Power BI:** Provides reporting dashboards and analytics for HR leadership.
- **Email/Calendar Systems:** Send automated notifications and reminders.
- **AWS Lambda:** Executes serverless functions to automate backend processes and event-driven tasks.

Additional assumptions include:

- All services operate within cloud-based environments and require active enterprise licenses.
- Users access the system through modern web browsers with internet connectivity.
- HR maintains accurate employee and supervisor information for proper workflow routing.

2.2. General Constraints

- System availability depends on Smartsheet, Box, and Power BI service uptime.
- API rate limits and vendor service restrictions may affect automation performance.
- Access to the system is limited to authorized County staff through Single Sign-On (SSO).
- Customization is limited to the capabilities provided by the supported enterprise platforms.

2.3. Goals and Guidelines

- Centralize HR workflow tracking for performance reviews, personnel matters, recruitment activities, and employee separations.
- Improve visibility and accountability across HR processes.
- Provide automated reminders and status tracking to ensure timely completion of tasks.
- Maintain secure document storage and reporting capabilities aligned with County policies.

2.4. Development Methods

An Agile development approach was followed. System features were implemented and tested through iterative development cycles, allowing continuous feedback from project sponsors and HR staff. Responsibilities were divided among team members to support parallel development and faster integration of system components.

3. Architectural Strategies

This section describes the major design strategies used in the Intelligent HR Analytics Platform, including technology selection, system reuse, integration methods, and scalability considerations.

3.1. Technology Stack Selection

- The system is built using cloud-based enterprise platforms including Smartsheet, Box, and Power BI.
- Smartsheet serves as the workflow engine for tracking Employee Performance Reviews (EPR), Personnel Matters, Vacancies and Recruitment activities, and Employee Separations.
- Box provides secure document storage and archiving for finalized HR records.
- Power BI generates dashboards and analytics reports for HR leadership and management.
- AWS Lambda supports serverless, event-driven automation and system integrations across the platform.
- These platforms were selected because they integrate with existing County infrastructure and allow workflow automation without extensive custom development.

3.2. Reuse of Existing Components

- The platform leverages existing enterprise tools already approved by the County rather than building new infrastructure from scratch.
- Smartsheet automation rules are reused across multiple workflows to standardize tracking and notifications.
- Existing Box storage policies are used to manage document retention and access control.

3.3. Integration and Communication

- Smartsheet integrates with Box to store finalized documents and maintain secure document links.
- Power BI connects to Smartsheet datasets to generate reports and dashboards.
- AWS Lambda facilitates event-driven integrations and automation between systems.
- Automated notifications and reminders are delivered through email.
- All integrations occur through secure APIs using HTTPS and OAuth authentication.

3.4. Modularity and Scalability

- The system follows a modular design where each workflow (EPR, Personnel Matters, Vacancies and Recruitment, and Separations) operates as an independent subsystem.
- Each subsystem maintains its own tracker sheets and automation rules while sharing the same architecture.
- Additional HR workflows can be added without major structural changes.

3.5. Data Storage and Management

- Smartsheet stores operational workflow data and status tracking information.
- Box stores finalized documents and case files.
- Power BI consumes structured datasets to provide reporting and analytics.

3.6. Security Strategy

- Role-Based Access Control (RBAC) restricts system access based on user role.
- All integrations use encrypted HTTPS connections and OAuth authentication.
- Audit logging is enabled to track workflow updates and document access.

3.7. Rejected Alternatives

The following architectural approaches were evaluated but not implemented:

- Manual PDF routing via email: lacked centralized tracking and analytics capability.
- DocuSign for Digital Signatures: Evaluated as a potential solution for document signing; however, it was not adopted due to certification and compliance limitations within the County environment.

4. System Architecture

The Intelligent HR Analytics Platform is implemented using a modular, SaaS-based distributed architecture. The system is decomposed into interoperable subsystems that communicate through authenticated API integrations and native platform connectors.

The architecture supports multiple HR workflow domains while maintaining a shared integration and security foundation.

4.1. System Decomposition

The system is divided into functional subsystems built on a shared integration framework.

4.1.1. Workflow Subsystems (Smartsheet-Based)

Each HR initiative is implemented as an independent workflow module within Smartsheet. All workflow modules share a consistent structural pattern:

- Dedicated tracking sheet(s)
- Structured column schema
- State-controlled Status field
- Automation rule groups
- Approval routing logic
- Archival trigger

EPR Workflow Subsystem Manages review creation, supervisor assignment, approval routing, digital acknowledgments, tracking completion status, and initiating archival.

Personnel Matters Subsystem Handles administrative record tracking, document management, optional metadata extraction, approval routing, and archival.

Vacancies and Recruitment Subsystem Creates and tracks recruitment records, manages hiring status and key steps.

Separations Subsystem Implements structured separation intake, validation logic, outbound communication tracking, status management, and archival initiation.

4.1.2. Secure Document Repository (Box.com)

Serves as the centralized storage layer for finalized HR documents. Responsibilities include:

- Structured folder hierarchy enforcement
- Role-based access control
- Retention policy enforcement
- Audit logging

4.1.3. Analytics and Reporting Subsystem (Power BI)

Provides cross-workflow dashboards and reporting capabilities. Responsibilities include:

- Dataset ingestion from Smartsheet
- Data modeling and normalization
- KPI computation
- Interactive dashboard presentation

4.1.4. Integration and Data Services Layer

Coordinates secure data flow between Smartsheet, Box.com, external APIs, and Power BI using REST APIs, OAuth-based authentication, and scheduled synchronization mechanisms.

4.2. Component Responsibilities

Each architectural component maintains defined boundaries and roles.

4.2.1. Smartsheet

- Operational workflow engine
- Primary system of record for structured HR metadata
- Automation rule execution
- Approval routing and status transitions
- Notification generation

4.2.2. Box.com

- Immutable document repository
- Structured folder organization
- Metadata tagging
- Access control enforcement
- Activity logging

4.2.3. Power BI

- Analytical data modeling
- KPI and metric computation
- Dashboard visualization
- Scheduled dataset refresh

4.2.4. Integration Layer

- API invocation and token handling
- Data validation and transformation (if required)
- Retry handling for transient failures
- Synchronization status logging

4.3. Communication and Control Flow

All workflow domains follow a standardized event-driven lifecycle model.

4.3.1. Workflow Lifecycle

1. **Initiation** — Triggered by form submission, row creation, or time-based automation.
2. **Processing** — Automation rules evaluate conditions and update state.
3. **Approval Routing** — Sequential or single-stage approvals executed where applicable.
4. **Completion Validation** — Required fields and approvals verified.
5. **Archival Trigger** — Finalized artifacts uploaded to Box; metadata written back.
6. **Analytics Refresh** — Power BI incorporates updated data during scheduled refresh.

Each subsystem adheres to this control model to ensure architectural consistency.

4.4. Data Flow and Integration Logic

The system primarily uses one-way data synchronization patterns to preserve data integrity.

4.4.1. Smartsheet → Box (Archival Flow)

- Trigger: Workflow reaches final completion state.
- Action: Upload finalized document with metadata.
- Write-back: Store Box file URL in Smartsheet.
- Direction: One-way.

4.4.2. Smartsheet → Power BI (Analytics Flow)

- Trigger: Scheduled dataset refresh.
- Action: Import structured workflow data.
- Direction: One-way read-only consumption.

4.4.3. External APIs → Smartsheet (Optional Metadata Extraction)

- Trigger: Document upload event or row update.
- Action: Extract structured fields from document.
- Write-back: Populate normalized Smartsheet columns.

4.4.4. External APIs → Separations (Email Notifications)

- Trigger: Separation request initiated or status updated.
- Action: Generate and send customized email notifications.
- Data Source: Smartsheet records and associated documents (Box).
- Delivery: Email system via secure API.

All integrations use HTTPS encryption and OAuth-based authentication.

4.5. Scalability and Modularity

4.5.1. Subsystem Isolation

Workflow domains operate independently using separate sheets and automation rule groups. Changes to one subsystem do not require structural changes to others.

4.5.2. Scalability Approach

The architecture supports scaling through:

- Increased row volume within Smartsheet trackers
- Expanded Box folder hierarchies
- Enlarged Power BI datasets

No architectural redesign is required for projected growth within expected operational bounds.

4.6. Justification of Architecture

The selected architecture satisfies the following objectives:

- **Scalability** — Configuration-based growth without backend expansion.
- **Reusability** — Shared integration and workflow patterns across domains.
- **Security** — OAuth-based authentication and role-based access control.
- **Maintainability** — Low-code automation reduces long-term technical debt.
- **Auditability** — Structured tracking and logging across all workflows.

5. Design Policies and Implementation Tactics

This section defines the technical standards, architectural policies, and implementation decisions governing the Intelligent HR Analytics Platform. These guidelines ensure consistency, maintainability, and security across all workflow domains.

5.1. Platform Components

5.1.1. Smartsheet (Workflow Engine)

Smartsheet functions as the operational workflow engine and system of record for structured HR workflow data. It manages workflow state, routing logic, and automation rules through controlled Status fields. Only structured metadata is stored within Smartsheet. Finalized document artifacts are stored externally, with references (e.g., Box_URL) maintained in workflow records.

5.1.2. Box.com (Document Repository)

Box provides secure storage for finalized workflow documents such as performance reviews, personnel files, recruitment artifacts, and separation documentation. Folder-level permissions and metadata tagging support controlled access, document retrieval, and retention policy enforcement.

5.1.3. Power BI (Analytics Engine)

Power BI provides analytics and reporting capabilities. Operational data from Smartsheet is modeled and transformed within Power BI to generate dashboards and performance metrics using DAX measures. Power BI operates in read-only mode with respect to workflow data.

5.1.4. Integration and Authentication

System integrations use secure HTTPS communication with OAuth 2.0 authentication. Access permissions follow the principle of least privilege, and credentials are never embedded directly in configuration files or scripts.

5.2. Requirements Traceability

Traceability between requirements and implementation is maintained through a Requirements Traceability Matrix (RTM). Each requirement maps to one or more implementation artifacts, including Smartsheet columns, automation rules, Box metadata fields, or Power BI report elements. Traceability is verified prior to major system releases.

5.3. Verification and Validation Strategy

System validation is performed at multiple levels to ensure functional correctness and data reliability.

5.3.1. Unit-Level Validation

Individual configuration elements are tested using sample data. Automation rules, column formulas, and Power BI measures are verified to ensure expected behavior and accurate calculations.

5.3.2. Testing

- **Unit Testing:** Individual components and workflows were tested to ensure correct functionality (e.g., form submissions, data updates, and automation logic).
- **Integration Testing:** System integrations between Smartsheet, Box, Power BI, and external APIs were tested to ensure proper data flow and communication.
- **Performance Testing:** Basic validation was conducted to ensure the system performs reliably under expected usage conditions.

5.4. Error Handling and Exception Management

- API failures are logged and retried where applicable.
- Data validation errors are flagged within Smartsheet workflows.
- Notification failures (e.g., email delivery issues) are monitored and logged.

5.5. Logging and Monitoring

- System activities and API interactions are logged for auditing and troubleshooting.
- Workflow execution and automation processes are monitored for failures.
- Logs are reviewed periodically to ensure system reliability.

5.6. Backup and Recovery

- Data stored in Box and Smartsheet relies on platform-managed backups.
- Critical records can be restored based on platform retention policies.
- Periodic data exports may be used for additional backup assurance.

5.7. Access Control

- Role-based access control is supported across Smartsheet, Box, and Power BI.
- User roles and permissions (e.g., HR staff, supervisors, administrators) will be configured based on organizational policies.
- Final access control settings will be defined and adjusted during system handoff to the sponsoring organization to align with internal security requirements.

5.7.1. Accessibility Validation

Dashboards and user interfaces are evaluated for accessibility, including appropriate color contrast, keyboard navigation support, screen-reader compatibility, and accessible export formats.

5.7.2. Data Integrity Monitoring

Data consistency between Smartsheet and Power BI is monitored through periodic comparisons of row counts and aggregate metrics. Significant discrepancies trigger manual review.

5.8. Engineering Trade-Off Decisions

5.8.1. Low-Code vs. Custom Development

The system prioritizes low-code platform capabilities to reduce infrastructure complexity and simplify long-term maintenance. Custom scripting is used only where required for integration or specialized validation.

5.8.2. Analytics Refresh Strategy

Analytics data is refreshed on a scheduled basis rather than in real time. Outside of work hours Power BI refresh cycles provide sufficient reporting accuracy while reducing API usage and operational overhead.

5.8.3. Source-of-Truth Model

The system maintains clear ownership of data across subsystems:

- Smartsheet — operational workflow data
- Box.com — finalized document repository
- Power BI — analytics and reporting

This separation prevents conflicting updates and maintains clear system responsibilities.

5.9. Design Standards

5.9.1. Date Handling

All system dates follow the ISO 8601 standard in UTC. Localization occurs only within presentation layers such as Power BI dashboards.

5.9.2. Box Metadata Policy

Archived documents must include metadata identifying the associated workflow record (e.g., EmployeeID or VacancyID), workflow domain, and retention classification. These metadata fields support auditability and retrieval.

5.10. Communication Protocols

All system communications occur over HTTPS using TLS 1.2 or higher. Integrations use OAuth 2.0 authentication and vendor-supported APIs or connectors. Audit logging is maintained according to county policy.

5.11. Architectural Patterns

The system architecture follows several design principles:

- **Event-driven workflows** triggered by status changes, dates, or form submissions.
- **Idempotent processing** to prevent duplicate actions or artifacts.
- **Retry mechanisms** for handling transient integration failures.
- **Separation of concerns** between workflow management, document storage, and analytics.

5.12. Abstraction Strategy

External systems are accessed through APIs and platform connectors, allowing workflow logic to remain independent of vendor-specific implementations. In Power BI, reusable DAX measures encapsulate business logic, simplifying report maintenance and ensuring consistency across dashboards.

6. Detailed System Design

All workflow domains follow a consistent structural pattern. Each workflow is implemented as a Smartsheet-based tracker with a controlled lifecycle managed through the Status field. Automation rules handle routing, notifications, and validation, while completed records trigger document archival in Box and reporting updates in Power BI.

- Structured Smartsheet tracker
- State-based workflow management via Status
- Automation rule groups for routing and notifications
- Archival trigger to Box
- Reporting integration with Power BI

6.1. EPR Workflow

Purpose Manages the full lifecycle of Employee Performance Reviews, including creation, routing through supervisory approvals, archival of finalized documents, and reset for future review cycles.

Core Data Fields EmployeeNumber, FirstName, LastName, JobClass, StartDate, EmploymentStatus, ProbationDueDate, SignedEPRDueDate, PreviousEPRSigned, PreviousEPRHR, Supervisor1-3, HeadHR, HeadOffice, Status, Box_URL

Workflow States

Not Created → With HR → With Supervisor1 → With Supervisor2
→ With Supervisor3 → With Head of HR → With Head of Office
→ Saving to Box → Completed

Integration Points

- Python-based automation for archival validation
- Box for secure document upload and storage
- Power BI for dataset refresh and reporting analytics

6.2. Personnel Matters Workflow

Purpose Tracks personnel-related administrative cases that require documentation, review, and archival.

Core Data Fields MatterID, OpenDate, Status, Complaint, Responder, IncidentSummary, IssueType, PolicyReference, Supervisor, NextStep, NextStepDueDate, Box_URL

Workflow States

Opened → In Progress → Closed

Integration Points

6.3. Vacancies and Recruitment Workflow

Purpose Tracks recruitment requests and hiring progress from vacancy creation to final position fill.

Core Data Fields VacancyID, JobClassTitle, Department, Label, VacancyStartDate, InterviewsConducted

Workflow States

Posted → Screening → Interviewing
→ Offer Extended → Offer Accepted → Filled → OnHold

Integration Points

- Optional document extraction via Box webhook and Python-based automation
- Power BI analytics and recruitment metrics

6.4. Separations Workflow

Purpose Tracks employee separation cases and manages communication workflows related to employee departures.

Core Data Fields EmployeeID, FirstName, LastName, Email, ExitReason, EffectiveDate, LastDayDate, EmailStatus

Workflow States

Missing Field → Awaiting Email → Email Sent → Email Failed

Integration Points

- Automated email generation and delivery
- Optional document processing and extraction
- Power BI reporting integration
- Optional document metadata extraction via Python-based automation
- Archival to Box repository
- Power BI reporting integration

7. Execution and Automation Model

This section defines the end-to-end automation architecture, including Smartsheet-native workflows and the external execution framework powered by AWS and Python.

7.1. Workflow Automation Structure

All workflows originate from Smartsheet and Box events, triggered via webhook integrations.

Core characteristics include:

- Event-driven execution via Smartsheet and Box webhooks
- Status-based lifecycle management within tracker sheets
- Conditional logic enforced through control and validation fields
- Idempotent processing using flags to prevent duplicate execution

Each tracker (e.g., Separation, EPR, Personnel Matters) follows a consistent structural design, while incorporating workflow-specific fields tailored to each process.

Fields are generally organized into three categories:

- **Core Fields:** Identify and track the record, including status, ownership, and key dates
- **Control Fields:** Support workflow execution through approvals, processing flags, and next-step indicators
- **Audit Fields:** Capture system activity, including timestamps, processing metrics, and execution logs

Specific field definitions vary by tracker to support the requirements of each workflow.

Smartsheet automation rules handle:

- Record initialization from intake forms
- Status transitions and approvals
- Notifications and reminders
- Flagging records for downstream processing (e.g., archival, sync)

7.2. Event-Driven Integration Architecture

External processing is initiated via webhook events from:

- Smartsheet (row creation/updates)
- Box (file/folder events)

These events are routed through AWS API Gateway, which directs requests to workflow-specific Lambda functions:

- EPR Lambda
- Personnel Matters Lambda
- Separations Lambda
- Vacancies & Recruitment Lambda

Each Lambda function invokes a corresponding Python script responsible for executing business logic.

7.3. Python Execution Framework

Python scripts perform operations not supported natively within Smartsheet, including:

- Box folder creation and archival synchronization
- Cross-system data validation and enrichment
- Metadata extraction and transformation
- Dynamic email content generation
- Bulk and asynchronous record processing

7.3.1. Execution Pattern

Each Lambda-triggered Python process follows a consistent pattern:

1. Retrieve event payload from API Gateway
2. Authenticate using credentials from AWS Secrets Manager
3. Identify and retrieve the relevant Smartsheet record(s)
4. Validate required fields and processing conditions
5. Execute workflow-specific actions (e.g., Box operations, updates)
6. Write results back to Smartsheet (status updates, timestamps)
7. Log execution outcomes for auditability

Processing is designed to be idempotent, ensuring repeated events do not create duplicate artifacts.

7.3.2. Security Controls

- Centralized credential management via AWS Secrets Manager
- OAuth 2.0 authentication for Smartsheet and Box APIs
- Least-privilege access for all services and scripts
- Structured logging within Lambda for traceability and monitoring

7.4. Performance Characteristics

- Fully event-driven architecture with no manual triggers
- Serverless execution via AWS Lambda enables horizontal scaling
- Asynchronous processing prevents blocking within Smartsheet workflows
- Designed to operate within API rate limits for all integrated systems

8. Database Design

This section describes the structure, purpose, and relationships of the primary Smartsheet data repository used by the Intelligent HR Analytics & EPR Management Dashboard. A single Smartsheet table (Employees) stores employee details, job classifications, and Employee Performance Review (EPR) workflow status. This centralized design simplifies automation, reporting, and integration with Box, and Power BI.

8.1. Data Integration

- Integrated with Power BI for reporting.
- Box stores completed EPR documents.
- Power Automate manages data synchronization across systems.

8.2. Security

- Role-based access (HR Admin, Supervisor, Employee).
- All changes are tracked for audit purposes.
- Secure API access via OAuth 2.0.

8.3. Data Tables

8.3.1. EPR Tracker - Employees Performance Tracker

Purpose: Stores employee performance review (EPR) data, including employee details, review timelines, and multi-level approval workflow.

Key Fields (Grouped):

- **Employee Info:** Employee Number, Name, Job Class, Level, Step, Start Date
- **EPR Details:** Anniversary Month, Employment Status, Probation Info, Due Dates
- **Tracking:** Status, Late EPR, Signed EPR Dates, Previous EPR Records
- **Assignments:** Supervisor(s), HR, Head of HR, Head of Office
- **Approvals:** Supervisor, HR, and Head of Office approval statuses
- **System:** Reset Ready flag, internal reference fields

Constraints:

- Only authorized roles can update approval and status fields.

	Status	Docusi... Status	Primary Column	Employee Number	Employee First Name	Employee Last Name	Job Class	Job Class Level	Job Class Step	Start Date	Anniver... Month	Probation... EPR	Employment Status	Probation Quarter	Probation Due Date	Late EPR	Signed EPR Due Date	Previous EPR Signed	Previous EPR Actual Due Date
1	With Head of Offi		1	453729	James	San	ADMHS Case Work	Senior	A	03/18/20	Jan		yearly	N/A	09/26/25		11/28/25	11/28/24	11/28/24
2	With Head of Offi		2	105864	Isabella	Smith	ADMHS Recovery A	N/A	C	11/07/19	Apr		probationary	1Q	01/09/25		11/21/25	11/21/24	11/21/24
3	With Head of Offi		4	927351	Emma	Brown	HR Manager	I	F	06/25/18	Aug		probationary	2Q	10/26/23		11/01/25	11/01/24	11/01/24

Figure 1: EPR 1

	Late EPR	Signed EPR Due Date	Previous EPR Signed	Previous EPR Actual Due Date	Supervisor	HR	Supervisor 1	Supervisor 2	Supervisor 3	Head of HR	Head of Office	Supervisor 2 approval	Supervisor 3 approval	Head of HR approval	Head of Office approval	Reset Ready
1		11/28/25	11/28/24	11/28/24	Joe Lopez							Approved	Approved	Approved	Submitted	
2		11/21/25	11/21/24	11/21/24	Adrian Galvan							Approved	Approved	Approved	Submitted	
3		11/01/25	11/01/24	11/01/24	Callie Steele							Approved	Approved	Approved	Submitted	

Figure 2: EPR 2

8.3.2. EPR Tracker - Employees History Records

Purpose: Tracks the processing status of EPR records, including document handling (e.g., DocuSign), review progress, and approval workflow.

Key Fields (Grouped):

- **Processing Info:** Date Saved, Status, DocuSign Status, Primary Column
- **Employee Info:** Employee Name, Job Class, Level, Step
- **EPR Details:** Anniversary Month, Probation Info, Due Dates, Late EPR
- **Assignments:** Supervisor(s), HR, Head of HR, Head of Office
- **Approvals:** Supervisor, HR, and Head of Office approval statuses
- **Tracking:** Previous EPR dates, Employment Status, Reset Ready flag

Constraints:

- Only authorized roles can update status and approval fields.

	Date Saved	Status	DocuSign Status	Primary Column	Employee First Name	Employee Last Name	Job Class	Job Class Level	Job Class Step	Anniversary Month	Probationary EPR	Probation Quarter	Probation Due Date	Late EPR	Signed EPR Due Date
1	11/14/25	Saving to Box	Not Started	1	James	San	Assessment Public Defn	Senior	A	Jan	☐	Final	09/30/25	☐	11/19/25
2	11/14/25	Saving to Box	Not Started	2	Isabella	Smith	ADMINIS Recovery A...	N/A	C	Apr	☐	Final	01/09/25	☐	11/19/25
3	11/20/25	Saving to Box	Not Started	2	Isabella	Smith	ADMINIS Recovery A...	N/A	C	Apr	☐	N/A	01/09/25	☐	11/19/25

Figure 3: EPR History 1

	Previous EPR Signed	Previous EPR Actual Due Date	Supervisor	HR	Supervisor 1	Supervisor 2	Supervisor 3	Head of HR	Head of Office	Supervisor 2 approval	Supervisor 3 approval	Head of HR approval	Head of Office approval	Reset Ready	Employment Status	Previous EPR Signed
1	09/18/24	09/18/24	Joe Lopez	picked 876	James San	James San	James San	James San	James San	Approved	Approved	Approved	Approved	☐		
2	01/11/25	01/11/25	Adrian Gekhan	picked 876	James San	James San	James San	James San	James San	Approved	Approved	Approved	Approved	☐		
3	01/11/25	01/11/25	Adrian Gekhan	picked 876	James San	James San	James San	James San	James San	Approved	Approved	Approved	Approved	☐		
4		11/29/24	Joe Lopez	picked 876	James San	James San	James San	James San	James San	Approved	Approved	Approved	Approved	☐	yearly	11/29/24

Figure 4: EPR History 2

8.3.3. Personnel Matters - Personnel Matters Tracker

Purpose: Tracks employee relations cases, including complaints, investigation details, status, and outcomes.

Key Fields (Grouped):

- **Case Info:** Matter ID, Opened Date, Status, Days Open
- **Complaint Details:** Complaint Type, Issue, Policy, Description of Incident
- **People Involved:** Respondent, Supervisor, Intake Submitted Via
- **Case Management:** Next Steps, Due Dates, Memo Notes
- **Outcome & Closure:** Closed Date, Outcome, Missing Close Info
- **Tracking:** Modified By, Last Modified Date, Fiscal Year, Update Required

Constraints:

- Sensitive fields are restricted to authorized HR personnel.

	Matter IDs	Opened Date	Status	Complaint(s)	Responde...	What Happened	Issue	Policy	Intake Submitt... Via	Supervi...	Next Steps	Next Step Due Date	Memo Notes	Closed Date	Outcome	Modified By	Last Modified Date	Fiscal Year	Upd... Requ...	Days Open	Missing Close Info
1	PM-001	09/21/25	Closed	Conduct	Isabella Smith		Conduct	Department Policies & Proc	HR	Shawna Mateer				10/19/25	Policy reminder	HR Rep A	10/23/25	FY25-26			
2	PM-002	05/24/25	Opened	Attendance	Isabella Miller	Late arrivals.	Attendance	Department Policies & Proc	HR	Erica Bortoff		09/02/25	Notes added by HR.			HR Rep A	09/07/25	FY25-26		204	
3	PM-003	12/09/25	In Progress	Workplace Conflict	Ethan Garcia	Employee reported experiencing unwelcome behavior from a coworker.	Workplace Conflict	SBCO Civil & Respectful W	Employee	Deepak Budwani						HR Rep C	12/17/25	FY25-26		97	

Figure 5: Personnel Matters Data

8.3.4. Separations - Employees Separations Tracker

Purpose: Tracks employee separation records, including exit details, payroll timelines, and notification status.

Key Fields (Grouped):

- **Submission Info:** Date Form Submitted, Separation ID, Email Status
- **Employee Info:** Employee ID, Name, Staff Email, Position ID, Job Class, Level
- **Separation Details:** Reason for Exit, Last Day Date
- **Payroll Tracking:** Pay Period, Payroll Start/End Dates, Final Pay Date
- **Benefits:** Insurance Last Date

Constraints:

- Sensitive employee data is restricted to authorized HR personnel.

	Date Form Submitted	Separati... ID	Email Status	Empl... ID	Employee First Name	Employee Last Name	Staff Email	Position ID	Job Class	Job Class Level	Reason for Exit	Last Day Date	Separated Pay Period	Separated Payroll Starting Date	Separated Payroll Ending Date	Separated Final Pay Date	Insurance Last Date
1	02/11/26 9:24 A	SEP-001	email sent	11111	Snoopy	Dog	James San	00001	Not Yet Assigned	III	Extra Help assign	01/18/26		01/05/26	01/18/26	01/29/26	01/31/26
2	02/11/26 9:19 A	SEP-002	email sent	22222	Charlie	Brown	James San	00002	Not Yet Assigned	Intern	Failed probation	09/17/25		09/15/25	09/28/25	10/09/25	09/30/25
3	02/11/26 9:24 A	SEP-003	missing fields	33333	Thomas	Shelby	Isack076	00003	Not Yet Assigned	Senior	Misconduct	02/25/26		02/16/26	03/01/26	03/12/26	03/31/26
4	02/18/26 4:15 P	SEP-004	email sent	10586	Isabella	Smith	Momoka Aur	00004	ADMHRS Recovery Assistant	III	Inadequate perfo	12/31/24		12/23/24	01/05/25	01/16/25	01/31/25
5	02/18/26 4:15 P	SEP-005	email sent	92735	Erinna	Brown	Momoka Aur	00002	HR Manager	Intern	Misconduct	02/25/25		02/17/25	03/02/25	03/13/25	03/31/25
6	02/18/26 4:15 P	SEP-006	email sent	64021	Olivia	Miller	Momoka Aur	00003	Public Defender CFAO	Senior	Inadequate perfo	02/25/24		02/19/24	03/03/24	03/14/24	03/31/24

Figure 6: Separation Data

8.3.5. Separations - Recognized Holidays

Purpose: Maintains a list of organizational holidays, including dates and rules for recurring holidays.

Key Fields (Grouped):

- **Holiday Info:** Holiday ID, Holiday Name
- **Dates:** Previous Holiday Date, Upcoming Holiday Date
- **Recurrence Rules:** Fixed Day Every Year, Fixed Day, Fixed Week of Month

Constraints:

- Recurring holiday fields must be consistently defined for accurate date calculation.

	Holiday ID	Previous Holiday Date	Upcoming Holiday Date	Holiday Name	Fixed Day Every Year?	Fixed Day	Fixed Week of Month
1	HOLIDAY-01	01/01/26	01/01/27	New Year's Day	☐	N/A	N/A
2	HOLIDAY-02	01/19/26	01/18/27	M.L.King Jr.'s Birthday	☒	Monday	Third Week of Month
3	HOLIDAY-03	02/12/26	02/12/27	Lincoln's Birthday	☐	N/A	N/A
4	HOLIDAY-04	02/16/26	02/15/27	Washington's Birthday	☒	Monday	Third Week of Month
5	HOLIDAY-05	05/26/25	05/26/26	Memorial Day	☒	Monday	Last Week of Month
6	HOLIDAY-06	06/19/25	06/19/26	Juneteenth	☐	N/A	N/A
7	HOLIDAY-07	07/04/25	07/04/26	Independence Day	☐	N/A	N/A
8	HOLIDAY-08	09/01/25	09/07/26	Labor Day	☒	Monday	First Week of Month
9	HOLIDAY-09	10/13/25	10/12/26	Columbus Day	☒	Monday	Second Week of Month
10	HOLIDAY-10	11/11/25	11/11/26	Veteran's Day	☐	N/A	N/A
11	HOLIDAY-11	11/27/25	11/26/26	Thanksgiving Day	☒	Thursday	Fourth Week of Month
12	HOLIDAY-12	11/28/25	11/27/26	Thanksgiving Friday	☒	Friday	Fourth Week of Month
13	HOLIDAY-13	12/25/25	12/25/26	Christmas Day	☐	N/A	N/A

Figure 7: Separation Holiday Data

8.3.6. Vacancies and Recruitment - Vacancies and Recruitment

Purpose: Tracks job vacancies, recruitment progress, and hiring activity.

Key Fields (Grouped):

- **Vacancy Info:** Vacancy ID, Department, Position ID, Job Class Title
- **Status Tracking:** Status (e.g., Posted, Screening, Interviewing), Label
- **Timeline:** Vacancy Start Date, Date Filled, Vacancy Days Open
- **Recruitment Activity:** Recruitment Type, Interviews Conducted
- **System:** Row Created Timestamp

Constraints:

- Limited by SharePoint list structure.
- Recruitment status fields are updated by authorized HR personnel only.

	Row Created Timestamp	Vacancy ID	Dept	PosID	JobClassTitle	Status	Label	Vacancy Start Date	Date Filled	Vacancy Days Open	Recruitment Type	Interviews Conducted
1	02/21/26 1:25 AM	VAC-002	023	22911	COMPUTER SYSTEMS S	POSTED	FUNDED	02/20/26		24	INTERNAL	4
2	02/21/26 1:25 AM	VAC-003	023	25530	DEP PUBLIC DEFENDER	SCREENING	FUNDED	02/20/26		24	INTERNAL	1
3	02/21/26 1:25 AM	VAC-004	023	7518	DEP PUBLIC DEFENDER	INTERVIEWING	FUNDED	02/20/26		24	INTERNAL	2
4	02/21/26 1:25 AM	VAC-005	023	16009	DEP PUBLIC DEFENDER	OFFER EXTEND	FUNDED	02/20/26		24	INTERNAL	3
5	02/21/26 1:25 AM	VAC-006	023	22009	DEP PUBLIC DEFENDER	OFFER ACCEPT	FUNDED	02/20/26		24	EXTERNAL	1

Figure 8: Vacancies and Recruitments Data

This single-sheet data structure provides modularity, real-time synchronization, and high traceability while maintaining compliance with SBC PDO HR data governance policies.

9. User Interface

9.1. Interface Overview

The system does not implement a custom front-end application. User interaction occurs through three existing SaaS platforms:

9.1.1. EPR

Smartsheet Interface and Notifications

The EPR Tracker serves as the operational data source, storing HR-managed evaluation records submitted through intake forms, including Employee ID, Review Period, Status, Reviewer, ratings, comments, approvals, outcomes, and processing time.

FileAutomationFormsConnections

Employee Performance TrackerView onlyRequest to edit

PH...	Status	Deserv...	Em...	Employee	Employee	Job Class	Job Class	Job Class	Start	Review...	Employment	Probation...	Probation	Probation	Signed EPR	Previous	Previous	Late EPR	Supervisor
Col...		Status	No...	First Name	Last Name	Level	Step	Step	Date	Month	Status	EPR?	Quarter	Due Date	Due Date	EPR	EPR	Actual	
1	Not Created		48372	James	San	ADMINIS Case Work	Senior	A	03/16/20	Jan	yearly		N/A	08/26/20	04/11/27	04/04/20	04/11/20		Joe Lopez
2	Completed		48372	James	San	ADMINIS Case Work	Senior	A	03/16/20	Jan	yearly		N/A	08/26/20	04/11/28	03/27/20	04/11/20		Joe Lopez
3	With Head of Office		15886	Isabella	Smith	ADMINIS Recovery A	N/A	C	11/07/19	Apr	probationary		Q1	01/09/20	11/21/20	11/21/24	11/21/24		Adrian Gasken
4	With Head of Office		10722	Emma	Brown	HIS Manager	I	F	08/02/18	Aug	probationary		Q2	10/29/20	11/01/20	11/01/24	11/01/24		Cathy Steele
5	With Head of Office		64021	Olivia	Miller	Public Defender CDF	N/A	E	09/14/20	Dec	probationary		Q3	08/31/22	12/29/25	12/29/24	12/29/24		Deepest Buehse
6	With Head of Office		27842	Supina	Smith	Social Services Inter	N/A	E	01/02/19	Apr	probationary		Q4	05/02/21	10/29/25	10/29/24	10/29/24		Shanea Meezer
7	With Head of Office		21887	Michelle	Case	Public Defender Inva	N/A	N/A	12/02/18	Sep	fee probationary		Q1	03/04/23	12/28/25	12/28/24	12/28/24		Deepest Buehse
8	With HR		88124	Olivia	Miller	Financial Office Prof	I	C	04/27/19	June	yearly		N/A	06/30/24	03/04/25	03/04/25	03/04/25		Joe Lopez
9	With Supervisor 1		35087	Emma	Martinez	Public Defender Inva	I	E	08/15/18	Nov	yearly		N/A	04/22/24	01/12/26	11/21/25	06/01/26		Adrian Gasken
10	With Supervisor 1		27815	David	Taylor	HR Manager	II	B	02/11/20	Sep	yearly		N/A	12/28/24	12/14/25	01/12/25	06/03/25		Krista Redburn
11	With Supervisor 1		12880	Olivia	Williams	Assistant Public Def	N/A	E	10/30/19	Nov	yearly		N/A	01/04/23	03/12/26	03/12/25	03/12/25		Joe Lopez
12	With HR		87241	Emily	Jones	EPR Systems & Pro	II	C	05/06/18	Mar	yearly		N/A	01/19/22	01/12/26	01/25/25	01/25/25		Tanya Mueggs

Attachments

Row 2 Sheet 1

ActionsUpgraded...

04/14/2025, 10:17 PM...

Figure 9: EPR Smartsheet page

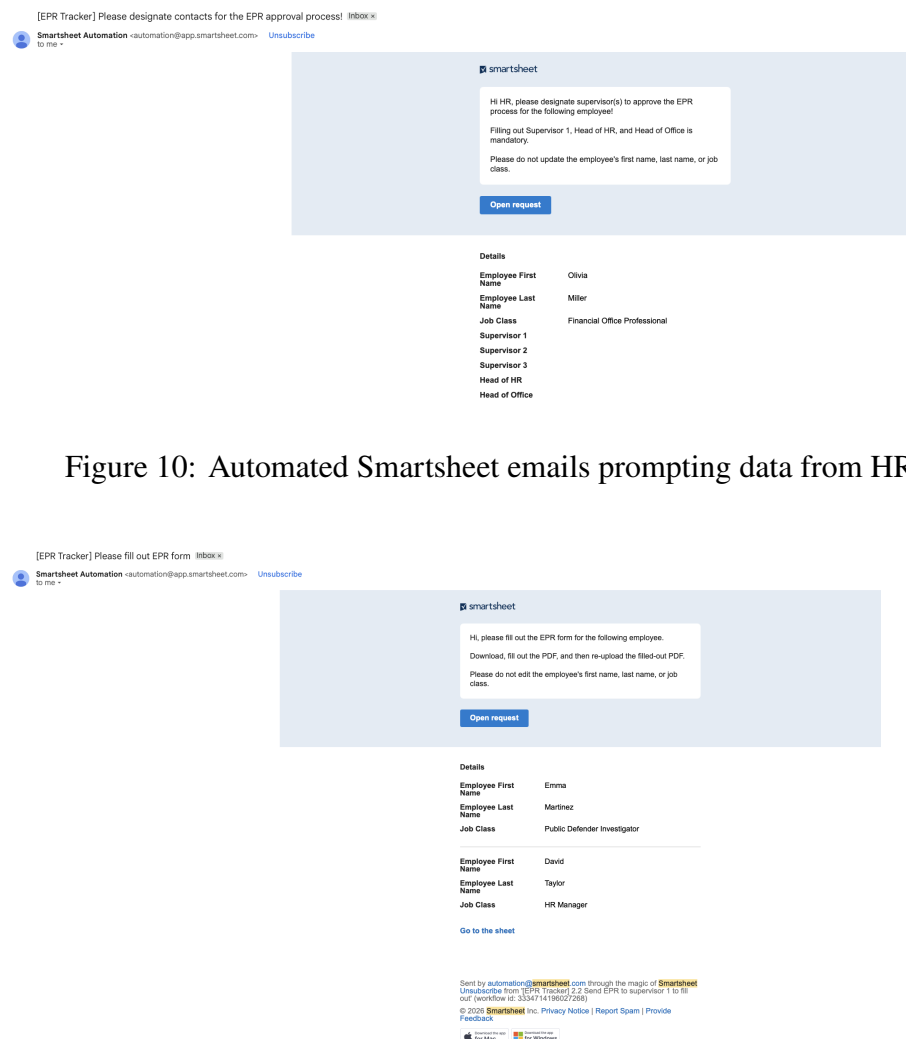


Figure 11: Smartsheet-generated email prompting user actions

Power BI Dashboard

The EPR Analytics Dashboard in Power BI provides near real-time insights into staffing and Employee Performance Review (EPR) progress. Data is refreshed nightly from Smartsheet.



Figure 12: PowerBI HR Dashboard

Page 1 – HR Overview Purpose: Provide a high-level view of workforce structure, job classifications, and reporting relationships.

Key Features:

- **Summary Cards (Employees, Supervisors, Interns, Employees on Probation)** Displays key workforce metrics that dynamically update based on filters and slicers.
- **Find Employee (Search Slicer)** Allows users to search and filter the dashboard by one or more employees.
- **Employee Directory** A table showing Full Name, Job Class, Job Class Level, and Supervisor. Selecting a row filters other visuals.
- **Employees by Job Class Level** Bar chart showing distribution across levels (Intern, I, II, III, Senior).
- **Employees by Job Class** Horizontal bar chart displaying employee counts by job category.
- **Employees by Job Class and Level (Stacked Bar Chart)** Combined visualization showing employee count by job class segmented by level.

Use: Supports workforce analysis, staffing distribution insights, and organizational planning.

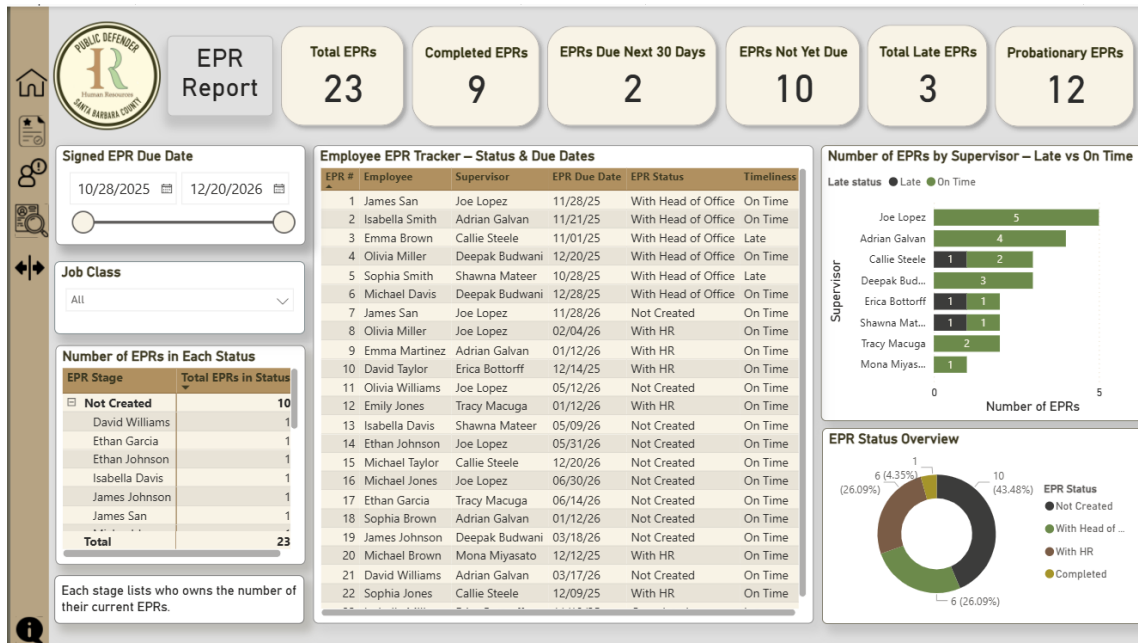


Figure 13: PowerBI EPR Dashboard

Page 2 – EPR Performance Review Purpose: Track the status and completion of Employee Performance Reviews.

Key Features:

- **Date Range Filter** Allows users to view EPRs within a selected time window.
- **KPI Cards** Displays Open EPRs, Completed EPRs, Total EPRs, and Completion Rate.
- **Open vs. Completed by Month** Line chart showing trends in review progress over time.
- **Open EPRs by Job Class** Bar chart highlighting outstanding reviews by job category.

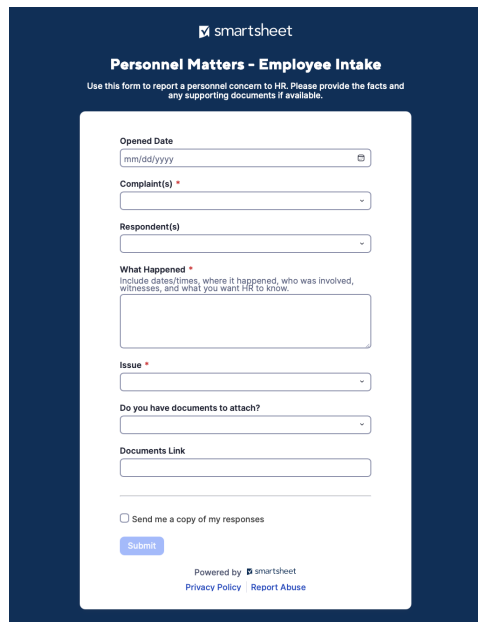
Key Metrics:

- Completion Rate (%) = Completed EPRs / Total EPRs
- Open EPRs = Reviews currently in progress
- Overdue EPRs = Reviews past due and not completed

Use: Helps HR and supervisors monitor review progress, identify overdue evaluations, and track completion performance.

9.1.2. Personnel Matters

Smartsheet / Intake Interfaces



Personnel Matters - Employee Intake

Use this form to report a personnel concern to HR. Please provide the facts and any supporting documents if available.

Opened Date
mm/dd/yyyy

Complaint(s) *

Respondent(s)

What Happened *
Include dates/times, where it happened, who was involved, witnesses, and what you want HR to know.

Issue *

Do you have documents to attach?

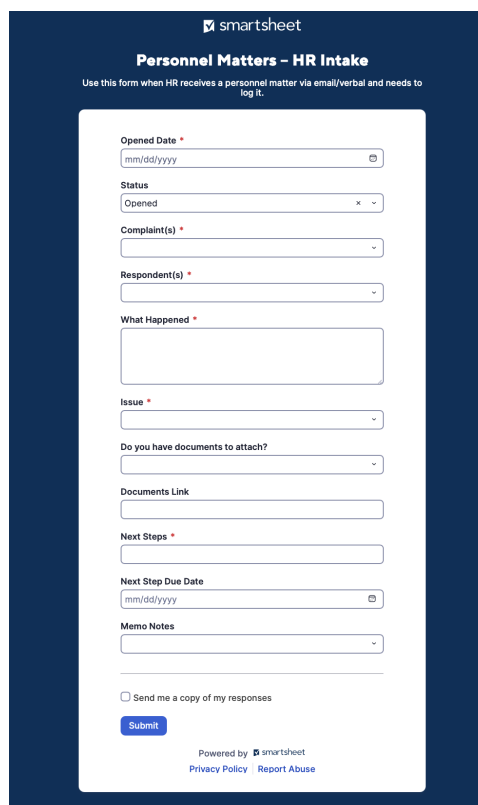
Documents Link

☐ Send me a copy of my responses

[Submit](#)

Powered by smartsheet
[Privacy Policy](#) [Report Abuse](#)

Figure 14: Personnel Matters Employee Intake Form



Personnel Matters - HR Intake

Use this form when HR receives a personnel matter via email/verbal and needs to log it.

Opened Date *
mm/dd/yyyy

Status
Opened

Complaint(s) *

Respondent(s) *

What Happened *

Issue *

Do you have documents to attach?

Documents Link

Next Steps *

Next Step Due Date
mm/dd/yyyy

Memo Notes

☐ Send me a copy of my responses

[Submit](#)

Powered by smartsheet
[Privacy Policy](#) [Report Abuse](#)

Figure 15: Personnel Matters HR Intake Form

Smartsheet / Data Tracker Interface

The Personnel Matters Smartsheet tracker serves as the operational data source for this component. It stores case-level records submitted through intake forms and maintained by HR, including fields such as Matter ID, Opened Date, Status, Complaint Type, Respondent, Issue, Supervisor, Next Steps, Closed Date, Outcome, Days Open, and update-required indicators. This

Matter IDs		Opened Date	Status	Complaint(s)	Respondent(s)	What Happened	Issue	Policy	Initial Supervisor	Supervisor	Next Steps	Next Step Due Date	Memo Notes	Closed Date	Outcome	Modified By	Last Modified Date	Fiscal Year	Update Required?	Days Open	Missing Close Info
1	PM-001	08/21/25	Closed	Conduct	Isabella Smith		Conduct issue		HR	Shawna Mateer				10/19/25	Policy reminder	HR Rep A	10/23/25	FY25-26			
2	PM-002	08/24/25	Opened	Attendance	Isabella Miller	Late arrivals	Attendance issue		HR	Erica Bottorff		08/02/25	Notes added by HR			HR Rep A	09/07/25	FY25-26		208	
3	PM-003	12/09/25	In Progress	Workplace Conflict	Ethan Garcia	Employee reported experiencing unwelcome behavior from a coworker	Workplace Conflict issue		Employee	Deepak Budhwani						HR Rep C	12/17/25	FY25-26		181	
4	PM-004	09/13/25	In Progress	Performance	Emily Jones	Supervisor reported employee is still struggling with a required task after training	Performance issue		HR	Shawna Mateer	Additional training and check-in were scheduled	10/06/25				HR Rep C	09/25/25	FY25-26		188	
5	PM-005	10/04/25	Opened	Workplace Conflict	James Johnson		Workplace Conflict issue		HR	Erica Bottorff			Notes added by HR			HR Rep B	10/04/25	FY25-26		167	
6	PM-006	10/28/25	In Progress	Attendance	Emma Brown		Attendance issue		HR	Tracy Macuga		11/13/25				HR Rep B	11/06/25	FY25-26		143	
7	PM-007	11/04/25	Closed	Performance	Ethan Johnson	Supervisor reported multiple missed deadlines	Performance issue		HR	Joe Lopez				12/23/25	Resolved internally	HR Rep B	11/13/25	FY25-26			
8	PM-008	02/02/26	In Progress	Conduct	Michael Jones		Conduct issue		HR	Erica Bottorff		03/24/26	Notes added by HR			HR Rep B	03/18/26	FY25-26		46	
9	PM-009	11/30/25	Closed	Conduct	Sophia Smith		Conduct issue		Employee	Callie Steele			Notes added by HR			HR Rep A	12/19/25	FY25-26			Missing Closed Date
10	PM-010	12/13/25	In Progress	Performance	Olivia Williams		Performance issue		HR	Tracy Macuga		01/03/26	Notes added by HR			HR Rep B	12/23/25	FY25-26		87	
11	PM-011	12/04/25	Opened	Attendance	Olivia Miller		Attendance issue		Employee	Deepak Budhwani		12/09/25	Notes added by HR			HR Rep A	12/14/25	FY25-26		106	
12	PM-012	10/23/25	In Progress	Safety	Ethan Johnson	Employee reported equipment-related safety concerns	Safety issue		HR	Shawna Mateer	Review details, schedule follow-up	11/03/25	Notes added by HR			HR Rep B	11/16/25	FY25-26		148	
13	PM-013	08/29/25	Opened	Harassment	Michael Taylor	Inappropriate comments made in office	Harassment issue		HR	Joe Lopez		09/12/25	Notes added by HR			HR Rep B	09/16/25	FY25-26		203	
14	PM-014	07/02/25	Opened	Workplace Conflict	Michael Davis		Workplace Conflict issue		Employee	Adrian Galvan		07/29/25	Notes added by HR			HR Rep C	07/05/25	FY25-26		281	
15	PM-015	10/29/25	Closed	Harassment	Isabella Davis		Harassment issue		HR	Deepak Budhwani				12/16/25	Separation/Resignation	HR Rep B	11/18/25	FY25-26			
16	PM-016	07/16/25	Opened	Policy	Emma Brown	Did not follow the required procedure	Policy issue		HR	Joe Lopez						HR Rep A	07/25/25	FY25-26		247	
17	PM-017	11/15/25	Opened	Conduct	Sophia Brown		Conduct issue		HR	Erica Bottorff		11/22/25	Notes added by HR			HR Rep B	12/08/25	FY25-26		125	
18	PM-018	10/30/25	Opened	Policy	Daniel Taylor		Policy issue		HR	Deepak Budhwani						HR Rep C	11/09/25	FY25-26		141	
19	PM-019	01/06/26	Opened	Attendance	Emily Jones		Attendance issue		HR	Joe Lopez			Notes added by HR			HR Rep B	01/15/26	FY25-26		73	
20	PM-020	12/04/25	In Progress	Performance	Isabella Smith		Performance issue		HR	Tracy Macuga		01/16/26	Notes added by HR			HR Rep A	01/15/26	FY25-26		86	
21	PM-021	02/06/26	Closed	Attendance	Daniel Taylor	Late arrivals	Attendance issue		Employee	Adrian Galvan		02/04/26			Received						

Figure 16: Personnel Matters Smartsheet Data Overview

tracker supports case monitoring, follow-up actions, and downstream reporting in the Power BI dashboard.

Power BI Dashboard

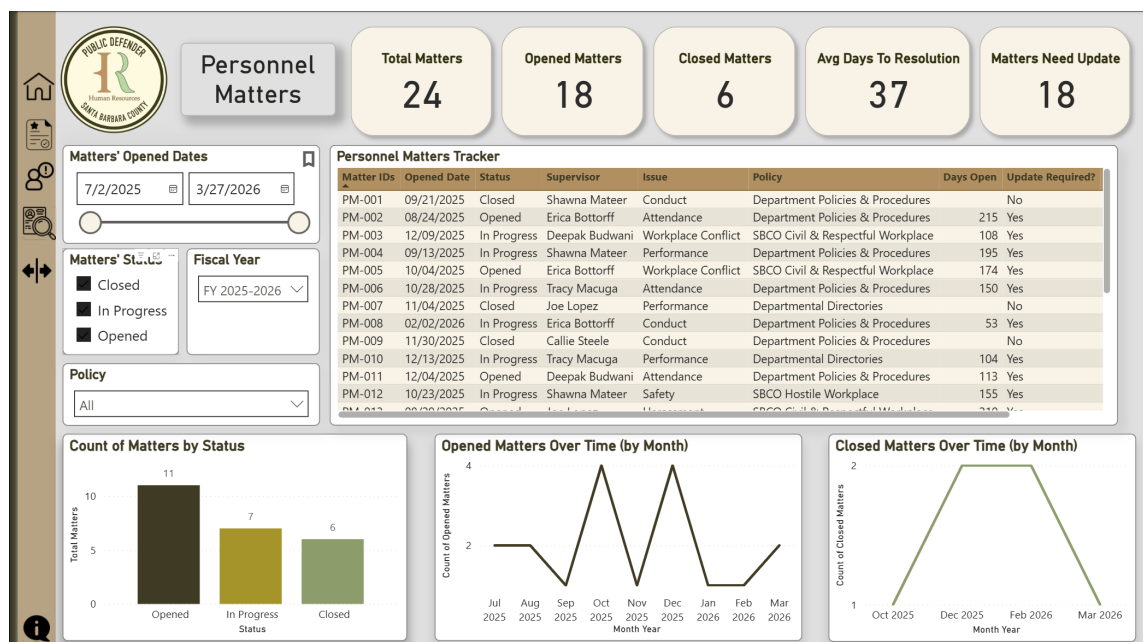


Figure 17: Power BI Personnel Matters Dashboard

The Personnel Matters Dashboard in Power BI provides near real-time visibility into employee relations cases, including open matters, closed matters, case status, and resolution activity. Data is refreshed from the Personnel Matters Smartsheet tracker and supports filtering for targeted HR review and compliance monitoring.

Overview Purpose: Provide HR and leadership with a centralized view of personnel matters so they can monitor case volume, case status, overdue follow-up, and resolution trends.

Key Features:

- **KPI Cards** display summary values such as Total Matters, Open Matters, Closed Matters, and Average Days to Resolution.

- **Status Breakdown** visual shows the distribution of cases by current status to help users quickly identify unresolved or in-progress matters.
- **Matters Opened Over Time** chart displays monthly or yearly intake trends for personnel matters.
- **Matters Closed Over Time** chart shows case resolution activity across time.
- **Detailed Tracker Table** displays case-level information such as Matter ID, Opened Date, Status, Complaint Type, Supervisor, and Closed Date.
- **Filter Controls** allow users to refine the dashboard by date range, fiscal year, status, complaint category, supervisor, or intake method.

Use: Helps HR monitor case progress, identify overdue or unresolved matters, support compliance reporting, and improve visibility into employee relations.

9.1.3. Vacancies and Recruitment

Smartsheet / Data Import Interface

The Vacancies and Recruitment component uses structured vacancy and recruitment data to support reporting on open positions, hiring progress, and workforce planning.

File Automation Forms Connections												
Vacancies and Recruitments View only ☆												
Grid Filter												
	Row Created Timestamp	Vacancy ID	Dept	PosID	JobClassTitle	Status	Label	Vacancy Start Date	Date Filled	Vacancy Days Open	Recruitment Type	Interviews Conducted
1	02/21/26 1:25 AM	VAC-002	023	22911	COMPUTER SYSTEMS SPEC I	POSTED	FUNDED	02/20/26		48	INTERNAL	4
2	02/21/26 1:25 AM	VAC-003	023	25530	DEP PUBLIC DEFENDER I	SCREENING	FUNDED	02/20/26		48	INTERNAL	1
3	02/21/26 1:25 AM	VAC-004	023	7518	DEP PUBLIC DEFENDER I	INTERVIEWING	FUNDED	02/20/26		48	INTERNAL	2
4	02/21/26 1:25 AM	VAC-005	023	16009	DEP PUBLIC DEFENDER I	OFFER EXTENDED	FUNDED	02/20/26		48	INTERNAL	3
5	02/21/26 1:25 AM	VAC-006	023	22098	DEP PUBLIC DEFENDER I	OFFER ACCEPTED	FUNDED	02/20/26		48	EXTERNAL	1
6	02/21/26 1:25 AM	VAC-007	023	25531	LEGAL OFFICE PRO I	FILLED	FUNDED	02/20/26	02/20/26	0	EXTERNAL	0
7	02/21/26 1:25 AM	VAC-008	023	25550	LEGAL OFFICE PRO I	ON HOLD	UNFUNDED	02/20/26		48	EXTERNAL	2
8	02/21/26 1:25 AM	VAC-009	023	23629	LEGAL OFFICE PRO I	ON HOLD	UNFUNDED	02/20/26		48	INTERNAL	3
9	02/21/26 1:25 AM	VAC-010	023	7495	FINANCIAL OFFICE PRO SR	FILLED	FUNDED	02/20/26	02/20/26	0	INTERNAL	0
10	02/21/26 1:25 AM	VAC-011	023	7497	LEGAL OFFICE PRO SR	FILLED	FUNDED	02/20/26	02/20/26	0	EXTERNAL	0
11	02/21/26 1:25 AM	VAC-012	023	22917	SOCIAL SERVICES WORKER	FILLED	FUNDED	02/20/26	02/20/26	0	EXTERNAL	0
12	03/06/26 9:11 AM	VAC-017	023	12345	LEGAL OFFICE PRO SR	POSTED		03/06/26		34		
13	03/06/26 9:11 AM	VAC-018	023	11112	DEP PUBLIC DEFENDER I	POSTED		03/06/26		34		
14												

Figure 18: Recruitment & Vacancies Smartsheet Data

Power BI Dashboard

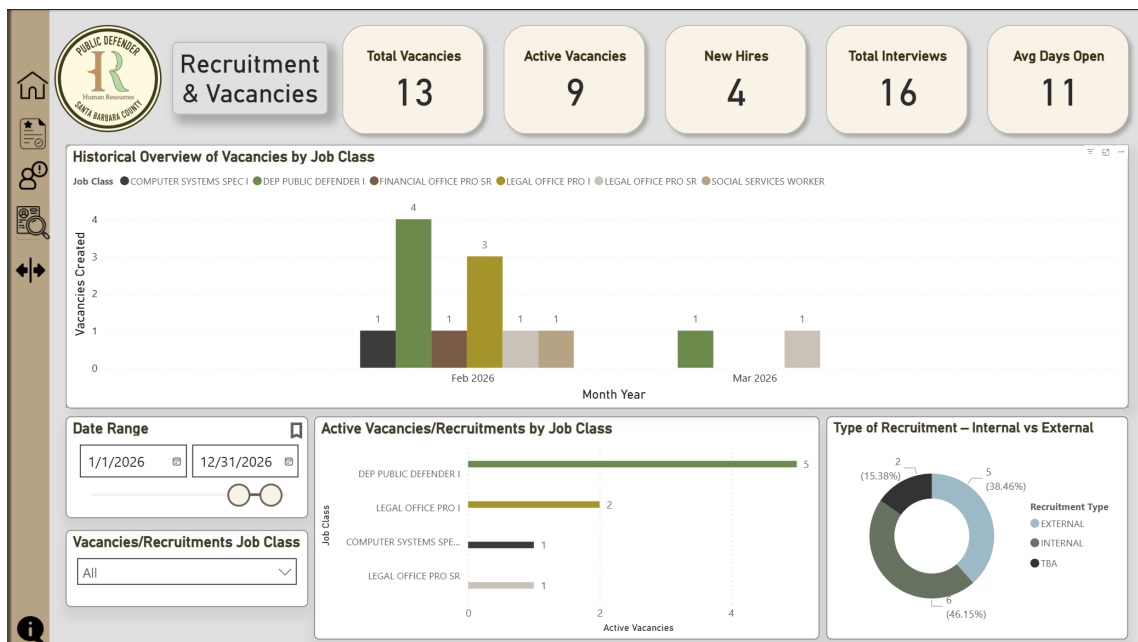


Figure 19: Power BI Vacancies Dashboard Page 1

Page 1 — Vacancies Overview Purpose: Provide a high-level view of current vacancy volume, active positions, and vacancy distribution across the organization.

Key Features:

- **KPI Cards** display summary values such as Total Vacancies, Active Vacancies, Filled Vacancies, and average vacancy days open.
- **Active Vacancies by Job Class** visual highlights which job categories currently have the highest staffing need.
- **Vacancy Status Breakdown** visual displays positions by workflow stage, such as Posted, Screening, Interviewing, Offer Extended, or Filled.
- **Vacancy Trend Over Time** chart shows how vacancy volume changes across months or fiscal periods.
- **Additional categorical visuals** may show vacancy distribution by funding label, department, office location, or other available organizational fields.
- **Filter controls** allow users to refine the dashboard by date range, fiscal year, job class title, department, funding label, or vacancy status.

Use: Supports staffing analysis, hiring prioritization, and workforce planning by helping HR identify where vacancies are concentrated and how long positions remain open.

Page 2 — Recruitment Overview Purpose: Track recruitment activity, hiring progress, and operational metrics related to filling positions.

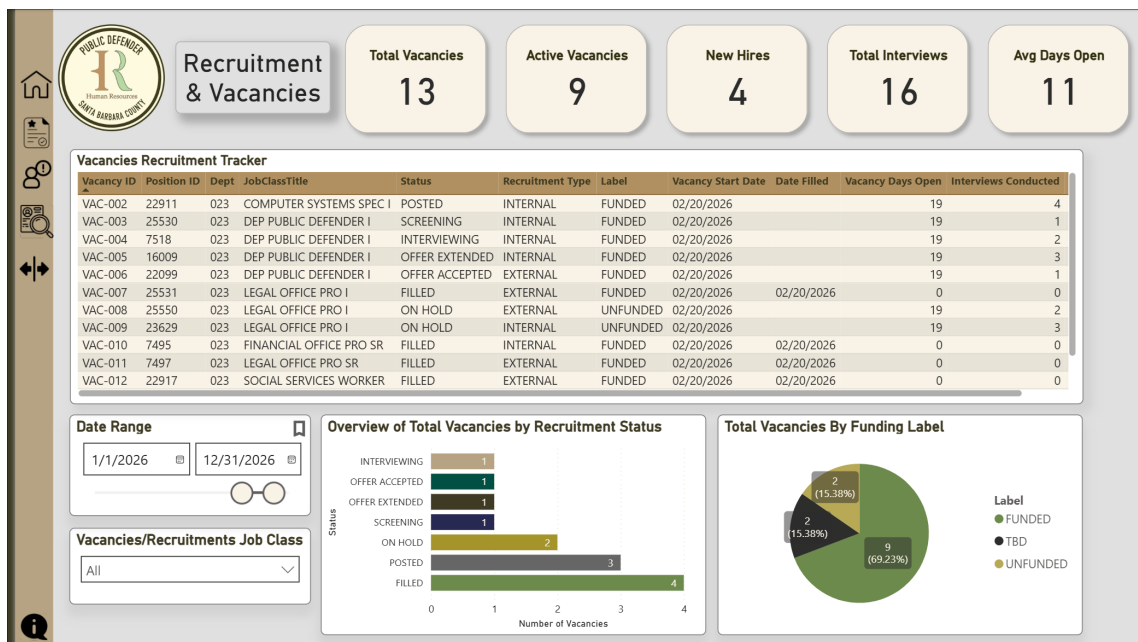


Figure 20: Power BI Vacancies Dashboard Page 2

Key Features:

- **KPI Cards** display summary values such as Recruitment Records, Interviews Conducted, New Hires, or other hiring activity metrics available in the dataset.
- **Recruitment Type Breakdown** visual shows internal versus external recruitment activity when applicable.
- **Hiring Activity Over Time** chart displays recruitment and hiring trends across selected time periods.
- **Historical Overview by Job Class** visual helps HR compare recruitment volume and outcomes across classifications.
- **Detailed Recruitment Table** provides row-level visibility into vacancy ID, job class title, recruitment type, interviews conducted, date filled, and current status.
- **Shared Filters** allow users to analyze recruitment data by job class, recruitment type, status, fiscal year, and date range.

Use: Helps HR track recruitment pipeline progress, monitor hiring outcomes, evaluate time-to-fill patterns, and support operational hiring decisions.

9.1.4. Separations

Smartsheet / Data Tracker Interface

The Separation Tracker serves as the operational data source, storing HR-managed case records submitted through intake forms, including Employee ID, Separation Type, dates, Status, Reason, Department, Supervisor, approvals, outcomes, and processing time.

	Date Form Submitted	Separati... ID	Email Status	Empl... ID	Employee First Name	Employee Last Name	Staff Email	Position ID	Job Class	Job Class Level	Reason for Exit	Last Day Date	Separated Pay Period	Separated Payroll Starting Date	Separated Payroll Ending Date	Separated Final Pay Date	Insurance Last Date
1.	02/11/26 9:24 A	SEP-001	email sent	11111	Snoopy	Dog	James San	00001	Not Yet Assigned	III	Extra Help assign	01/18/26		01/05/26	01/18/26	01/29/26	01/31/26
2.	02/11/26 9:19 A	SEP-002	email sent	22222	Charlie	Brown	James San	00002	Not Yet Assigned	Intern	Failed probation	09/17/25		09/15/25	09/28/25	10/09/25	09/30/25
3.	02/11/26 9:24 A	SEP-003	missing fields	33333	Thomas	Shelby	picok876	00003	Not Yet Assigned	Senior	Misconduct	02/25/26		02/16/26	03/01/26	03/12/26	03/31/26
4.	02/18/26 4:15 P	SEP-004	email sent	10586	Isabella	Smith	Momoka Aur	00004	ADMHS Recovery Assistar	III	Inadequate perfo	12/31/24		12/23/24	01/05/25	01/16/25	01/31/25
5.	02/18/26 4:15 P	SEP-005	email sent	92735	Emma	Brown	Momoka Aur	00002	HR Manager	Intern	Misconduct	02/25/25		02/17/25	03/02/25	03/13/25	03/31/25
6.	02/18/26 4:15 P	SEP-006	email sent	64021	Olivia	Miller	Momoka Aur	00003	Public Defender CFAO	Senior	Inadequate perfo	02/25/24		02/19/24	03/03/24	03/14/24	03/31/24

Figure 21: Separation Data

Figure 22: Separation HR Intake Form

Power BI Dashboard

The Separations Dashboard in Power BI provides insights into employee separations, including total exits, reasons for leaving, and trends over time. The dashboard supports filtering by job class, fiscal year, and date range to enable flexible analysis.

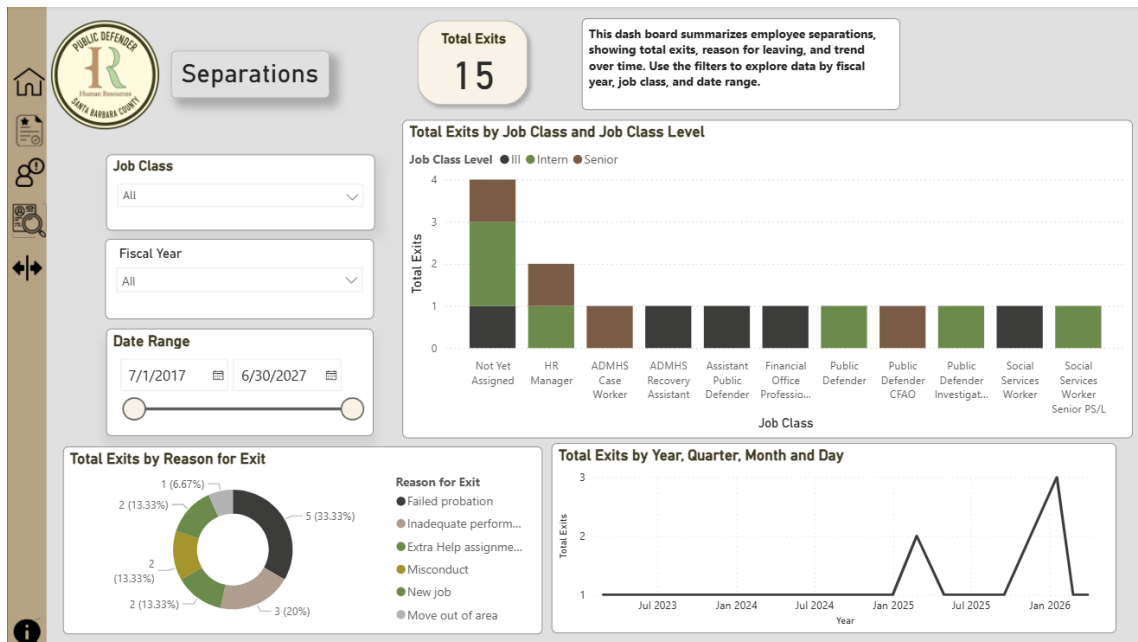


Figure 23: Separation PowerBI Dashboard

Overview Purpose: Provide a comprehensive view of employee separations to support workforce planning, retention analysis, and organizational decision-making.

Key Features:

- **Total Exits KPI** Displays the total number of employee separations within the selected filters.
- **Filters (Job Class, Fiscal Year, Date Range)** Allows users to refine the dataset dynamically for targeted analysis.
- **Total Exits by Job Class and Job Class Level** Stacked bar chart showing separations across job classes, segmented by level (e.g., Intern, I, Senior). *Use:* Helps identify which roles and levels experience the highest turnover.
- **Total Exits by Reason for Exit** Donut chart displaying the distribution of separation reasons (e.g., failed probation, new job, misconduct). *Use:* Supports understanding of primary drivers of employee turnover.
- **Total Exits Over Time** Line chart showing separation trends by year, quarter, month, and day. *Use:* Enables identification of patterns or spikes in employee exits over time.

Use: Supports HR in monitoring attrition, identifying turnover patterns, and informing retention strategies.

10. Requirements Validation and Verification

This section describes how key system functions of the Intelligent HR Analytics & EPR Management Dashboard are validated to ensure they meet defined requirements.

Requirement	Component / Module	Testing Method
Simple and accessible user interface for HR staff and supervisors	User Interface (Smartsheet, Power BI, Box)	Manual walkthroughs, user feedback sessions, and usability review
Ability to create, edit, and track employee performance reviews	Smartsheet Automation Rules	Functional testing by adding and updating records; verification of rule triggers
Correct routing of EPR forms for signatures	DocuSign Integration	Integration testing with sample forms; validation of signing order and completion notifications
Automatic storage of completed EPRs in Box	Box Storage Connector	File upload verification; validation of folder structure and access permissions
Display of EPR status and completion data in dashboards	Power BI Dashboard	Comparison of dashboard data with Smartsheet records; testing of filters and visuals
Daily update of Power BI datasets	Data Services Layer	Verification of scheduled refresh and dataset timestamps
Enforcement of user roles and permissions	Smartsheet / Box Permissions	Role-based access testing for HR, supervisors, and employees
Protection of data through secure access and encryption	Smartsheet, Box	Verification of HTTPS usage and compliance with County IT security policies
Support for multiple concurrent users without performance degradation	Smartsheet / Integration Layer	Simulation of concurrent updates and observation of system responsiveness
Recording of system activity for auditing purposes	Smartsheet History Log	Review of logs for accuracy of timestamps, user actions, and changes

All major modules were validated through user testing and workflow simulations to ensure the system operates correctly, securely, and efficiently.

11. Installation Guide

The following section will demonstrate the step-by-step process for setting up all the project scripts and webhook automations. This process involves many different technologies, such as the Python programming language, Smartsheet SDK, Box.com SDK, and many AWS Services, and different credentials from Smartsheet, Box.com, and AWS.

11.1. Prerequisites

All of the following must be completed before moving on to the following steps:

- Has administrative access to or a copy of the "SBPD - CSULA Spring 2026" Smartsheet workspace.
- Has administrative access or a copy of the "SBPD - CSULA" Box.com repository. If not, manually creating each folder and its subfolders will be required in step 11.2.
- Python version 3.11.4 installed locally.
- AWS SAM CLI installed locally.

11.2. Obtaining API Keys

11.2.1. Smartsheet

Obtaining a Smartsheet API key is simple. However, it is required to have Smartsheet's enterprise plan to generate an API key.

1. Navigate to <https://app.smartsheet.com/home>.
2. After signing in, click on your user icon on the bottom left corner.
3. Navigate to "Apps & Integrations...", then "API Access", and then click the "Generate new access token" button.
4. Create any name for the API key. Then save this string somewhere secure. This is your "Smartsheet API Key", we will be referencing it many times in this setup. Do not share this key with anyone!

11.2.2. Box.com

Obtaining a Box.com API key is slightly more difficult. However, Box.com allows you to generate an API key without any paid plan.

1. Navigate to <https://app.box.com/developers/console>.
2. After signing in, click on the "New app" button on the top right corner.
3. The "App Name" can be anything you want. However, select "Server Auth - JWT" for the App Type.
4. Scroll down to "Developer Actions" and enable "Manage webhooks".
5. Then click the "Generate Keypair" button to download the json file. Save this file somewhere secure on your desktop. This is your "Box.com API Credentials", we will be referencing it many times in this setup. Do not share this file with anyone!
6. Ensure Box.com folder structure follows the following schema:

```

SBPD - CSULA/
├── Personnel Matters/
├── Vacancies&Recruitments/
│   ├── 1` ADD NEW DEN FILE HERE!!/
│   ├── 2` USED DEN FILES - SAFE TO DELETE/
│   └── 3` INVALID DEN FILES - SAFE TO DELETE/
├── EPR Tracker/
│   └── EPR Attachments/
├── Separations/
│   ├── IMPORTANT - ATTACHMENTS SENT TO ALL EXITS/
│   └── email_template.boxnote

```

11.3. Setting up Developer Environment

In this section, we will be pulling the source code from github and then ensuring all the proper dependencies are downloaded.

Note: Use Python 3.11.4 as other versions may not work.

1. First, pull the source code from GitHub.

```
git pull https://github.com/PickolZi/CS4961-SBPD
```

2. Then, create and run a virtual environment to download the python dependencies.

```
python -m venv venv
```

```
source venv/Scripts/activate
```

3. Last, download all the python dependencies to the virtual environment.

```
pip install -r requirements.txt
```

11.4. Setting up Project Credentials

After the previous steps, all the internal CLI tools and scripts can run, however, they will fail due to credentials and configuration errors. In this section, we will add the credentials to the .env file to get half of the internal CLI tools working.

1. Rename the '.env.template' file to '.env'.
2. Set 'SMARTSHEET_ACCESS_TOKEN' to the Smartsheet API Key you saved from step 11.2.1
3. Using the JSON file you downloaded from Box.com during step 11.2.2, set the following values from the JSON file to the .env values.
 - BOX_CLIENT_ID
 - BOX_CLIENT_SECRET
 - BOX_JWT_KEY_ID
 - BOX_PRIVATE_KEY
 - BOX_PRIVATE_KEY_PASSPHRASE
 - BOX_ENTERPRISE_ID
4. Specifically for the Separations project, you can generate the Gmail credentials using the following link: <https://myaccount.google.com/apppasswords>. After generating the credentials, set both the 'GMAIL_SENDER_ADDRESS' and 'GMAIL_APP_PASSWORD'.

11.5. Setting up Project Configuration

The project configuration file `layers/shared/python/shared_config/config.py` is the centralized location for all project scripts to find which column to pull data from. And in order to figure out what the column ids are, we need to use an internal CLI tool we developed suited just for this purpose.

11.5.1. Filling in WebhookCLI Configuration class

1. Activate the `smartsheet_ids_cli.py` script and list all table IDs.

```
source venv/Scripts/activate
```

```
python smartsheet_ids_cli.py
```

> List Table IDs

```
(venv) C:\Users\picko\Documents\CS4961>python smartsheet_ids_cli.py
? Smartsheet IDs CLI List Table IDs
```

Sheet Name	Sheet ID	Sheet URL
[Personnel Matters] Tracker	1234840859922308	https://app.smartsheet.com/sheets/2rvG83rH8rhGvgRR3Cvgf6xxMh668HJfr3ppmVg1
[Personnel Matters] Tracker	4331645290106756	https://app.smartsheet.com/sheets/fMcqX7C8QC5vMG7G3vcQQQPC8mC3cJwXHmgmmg1
Copy of Employees	4559711836786564	https://app.smartsheet.com/sheets/VqvwqhvggPrg896VPCjhq6Vwgg9PPFjJJXWpxxV1
Employees	6605103877279620	https://app.smartsheet.com/sheets/gr2HqWgPwXMx9Pfr2JPRQV46gqxCXR3mRwqRMjj1
Employees History Records	7725880785719172	https://app.smartsheet.com/sheets/V2m4gX9fR3X9GV749xQ9GVWJfHQq5h7gWwjj2jvx1
Employees History Records	1945788230881156	https://app.smartsheet.com/sheets/PGPmfG9VcGGW9v3McWXPfRpj6Wf4fxFwRH5MxqG1
Employees Performance Tracker	2190844477001604	https://app.smartsheet.com/sheets/qcFQWH2fFvVH48WP7JfQrj9XWzMFx5F5xQJHhxjF1
Employees Separations Tracker	6507921459335044	https://app.smartsheet.com/sheets/cr49HR94P7xHvWC3cQ7RfC6fQWmcxv8qvpwqR21
Employees V3.0	2580213150994308	https://app.smartsheet.com/sheets/v8Cf8X2c2843GwJqQ94QCFpwm8hXvF4mHxcGCH91
james testing data	3595339459153796	https://app.smartsheet.com/sheets/V7R2Hp9w88chpmRvGFJwC89p3jrm4MGwxmmVRH71
Project Sheet	2499403039330180	https://app.smartsheet.com/sheets/FP86rRh9WMcFFR586p8Pfh3pV36R38f2xv4Gqjf1
Recognized Holidays	8351153952608132	https://app.smartsheet.com/sheets/pVH3r9phj78gXJ69xgP4jGjcq2w5jGwrxWJRJ2Qx1
Vacancies and Recruitments	4963448636002180	https://app.smartsheet.com/sheets/3RV39Q4wHxQJRQ49rjW5hH2fP9J5Ph3MGGPvHwM1
Vacancies/Recruitment	1039644897529732	https://app.smartsheet.com/sheets/m47Ww6jW386rjPfc4gM8HXVp8xhPhG6FRfQ9mMm1

Figure 24: `smartsheet_ids_cli.py` Listing Table IDs

2. Navigate to Box.com's folder titled "SBPD - CSULA / Vacancies&Recruitments / 1. ADD NEW DEN FILE HERE!!" And copy the folder's ID from the address bar.

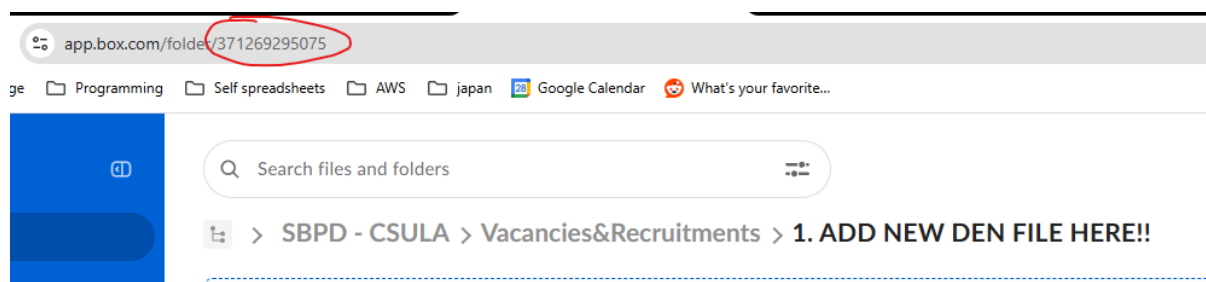


Figure 25: Vacancies & Recruitment Project - DEN Upload Folder ID

3. Overall, this is what the WebhookCLI class should look like.

```

14 class Config:
15     #####
16     # Fetching Smartsheet Table and Column IDs AND
17     # Toggling Webhooks for each project using AWS API Gateway & AWS Lambda
18     #####
19     class WebhookCli:
20         class Smartsheet:
21             EPR_TRACKER_TABLE_ID = 2190844477001604
22             PERSONNEL_MATTERS_TABLE_ID = 1234840859922308
23             SEPARATIONS_TRACKER_TABLE_ID = 6507921459335044
24             VACANCIES_AND_RECRUITMENT_TRACKER_TABLE_ID = 4963448636002180
25
26         class Box:
27             VACANCIES_DEN_UPLOAD_FOLDER_ID = 371269295075
28
29         class Aws:
30             EPR_TRACKER_API_GATEWAY_ADDRESS = os.getenv("EPR_TRACKER_API_GATEWAY_ADDRESS")
31             PERSONNEL_MATTERS_API_GATEWAY_ADDRESS = os.getenv("PERSONNEL_MATTERS_API_GATEWAY_ADDRESS")
32             SEPARATIONS_API_GATEWAY_ADDRESS = os.getenv("SEPARATIONS_API_GATEWAY_ADDRESS")
33             VACANCIES_API_GATEWAY_ADDRESS = os.getenv("VACANCIES_API_GATEWAY_ADDRESS")
34

```

Figure 26: Webhook CLI Class Config

11.5.2. Filling in EPRTracker Configuration class

1. Close and re-open the `smartsheet_ids_cli.py` script and then List EPR Tracker Column IDs.

```
python smartsheet_ids_cli.py
```

```
> List EPR Tracker Column IDs
```

```
(venv) C:\Users\picko\Documents\CS4961>python smartsheet_ids_cli.py
? Smartsheet IDs CLI List EPR Tracker Column IDs
Table: Employees Performance Tracker - 2190844477001604
Column Name | Column ID
-----+-----
Primary Column | 1895933145730948
Status | 3021833052573572
DocuSign Status | 7525432679944068
Employee Number | 1464870527258500
Employee First Name | 6399532773101444
Employee Last Name | 4147732959416196
Employee Full Name | 6947988347064196
Job Class | 8651332586786692
Job Class Level | 488558262177668
Job Class Step | 4992157889548164
Start Date | 5093893908156292
Anniversary Month | 2740358075862916
Employment Status | 1614458169020292
Probationary EPR? | 7243957703233412
Probation Quarter | 6118057796390788
Probation Due Date | 3866257982705540
Signed EPR Due Date | 1051508215598980
Previous EPR Signed | 5555107842969476
Previous EPR Actual Due Date | 3303308029284228
Late EPR | 8369857610076036
Supervisor | 7806907656654724
HR | 2177408122441604
Supervisor 1 | 6681007749812100
Supervisor 2 | 4429207936126852
Supervisor 3 | 8932807563497348
Head of HR | 136714541289348
Head of Office | 4640314168659844
Supervisor 2 approval | 2388514354974596
Supervisor 3 approval | 6892113982345092
Head of HR approval | 1262614448131972
Head of Office approval | 5766214075502468
```

Figure 27: smartsheet_ids_cli.py Listing EPRTracker column IDs

2. Overall, this is what the EPRTracker class should look like.


```

35 #####
36 # EPR Tracker Project
37 #####
38 class EPRTracker:
39     class Smartsheet:
40         EPR_TRACKER_TABLE_ID = 2190844477001604
41         EPR_TRACKER_HISTORY_TABLE_ID = 7725880785719172
42         STATUS_COLUMN_ID = 3021833052573572
43         FIRST_NAME_COLUMN_ID = 6399532773101444
44         LAST_NAME_COLUMN_ID = 4147732959416196
45         JOB_CLASS_COLUMN_ID = 8651332586786692
46         EMPLOYMENT_STATUS_COLUMN_ID = 1614458169020292
47         PROBATION_QUARTER_COLUMN_ID = 6118057796390788
48         SIGNED_EPR_DUE_DATE_COLUMN_ID = 1051508215598980
49         PREVIOUS_EPR_SIGNED_DATE_COLUMN_ID = 5555107842969476
50         PREVIOUS_EPR_ACTUAL_DUE_DATE_COLUMN_ID = 3303308029284228
51

```

Figure 28: EPRTracker Class Config

11.5.3. Filling in Separations Configuration class

1. List Separations Column IDs.

> List Separations Tracker Column IDs

? Smartsheet IDs CLI ~~List Separations Tracker Column IDs~~

Table: Employees ~~Separations Tracker - 6507921459335044~~

Column Name	Column ID
Date Form Submitted	741312352178052
Separation ID	7654345078951812
Email Status	3592840163315588
Employee ID	7621141206880132
Employee First Name	1991641672667012
Employee Last Name	4216701892382596
Staff Email	6495241300037508
Position ID	4243441486352260
Job Class	2946846820093828
Job Class Level	8747041113722756
Reason for Exit	162054324047748
Last Day Date	4169898010562436
Separated Pay Period	8030907343261572
Separated Payroll Starting Date	7363679878860676
Separated Payroll Ending Date	1734180344647556
Separated Final Pay Date	6237779972018052
Insurance Last Date	6412426227175300

Figure 29: smartsheet_ids_cli.py Listing Separations column IDs

2. Navigate to Box.com's Separations folder and attach both the email template boxnote file and the important attachments folder.

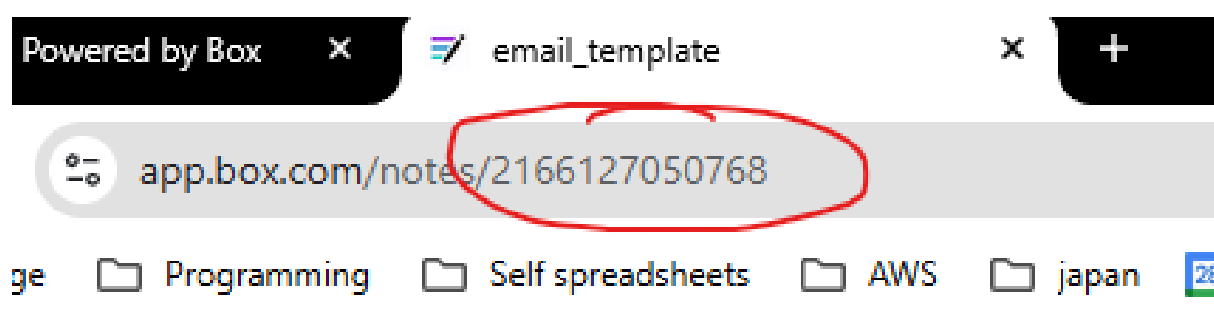


Figure 30: Separations Project - Box Email Template Boxnote File ID

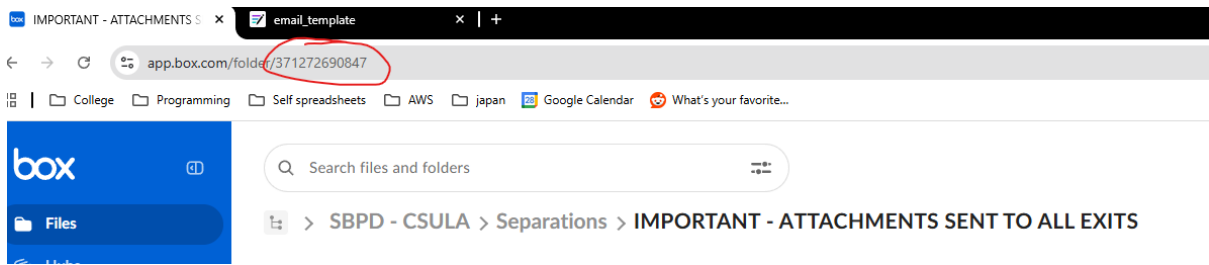


Figure 31: Separations Project - Box Important Attachments Folder ID

3. Overall, this is what the Separations class should look like.

```

52  #####
53  # Separations Project
54  #####
55  class Separations:
56      class Email:
57          SMTP_SERVER = "smtp.gmail.com"
58          PORT = 587
59          SENDER_ADDRESS = get_secret("GMAIL_SENDER_ADDRESS")
60          SENDER_APP_PASSWORD = get_secret("GMAIL_APP_PASSWORD")
61          SUBJECT = "SBPD - Separation Information email IMPORTANT!"
62
63      class Smartsheet:
64          SEPARATIONS_TRACKER_TABLE_ID = 6507921459335044
65          HOLIDAY_TABLE_ID = 8351153952608132
66          COLUMN_EMAIL_STATUS_ID = 3592840163315588
67          COLUMN_STAFF_EMAIL_COLUMN_ID = 6495241300037508
68          COLUMN_LAST_DAY_DATE_ID = 4169898010562436
69          HOLIDAY_PREVIOUS_DATES_COLUMN_ID = 4173747538579332
70          HOLIDAY_UPCOMING_DATES_COLUMN_ID = 4347590634852228
71
72      class Box:
73          IMPORTANT_ATTACHMENTS_TO_SEND_FOLDER_ID = 371272690847
74          EMAIL_TEMPLATE_FILE_ID = 2166127050768
75

```

Figure 32: Separations Class Config

11.5.4. Filling in Vacancies&Recruitment Configuration class

1. List Vacancies&Recruitment Column IDs.

> List Vacancies & Recruitment Tracker Column IDs

? Smartsheet IDs CLI List Vacancies & Recruitment Tracker Column IDs

Table: Vacancies and Recruitments - 4963448636002180

Column Name	Column ID
Row Created Timestamp	5939004221116292
Vacancy ID	7379808982617988
Dept	6773653411417988
PosID	5647753504575364
JobClassTitle	3395953690890116
Status	2833003737468804
Label	6210703457996676
Vacancy Start Date	581203923783556
Date Filled	1707103830626180
Vacancy Days Open	7336603364839300
Recruitment Type	5084803551154052
Interviews Conducted	3958903644311428

Figure 33: smartsheet_ids_cli.py Listing Vacancies & Recruitment column IDs

2. Navigate to Box.com's Vacancies&Recruitments folder and attach all 3 folder ids.

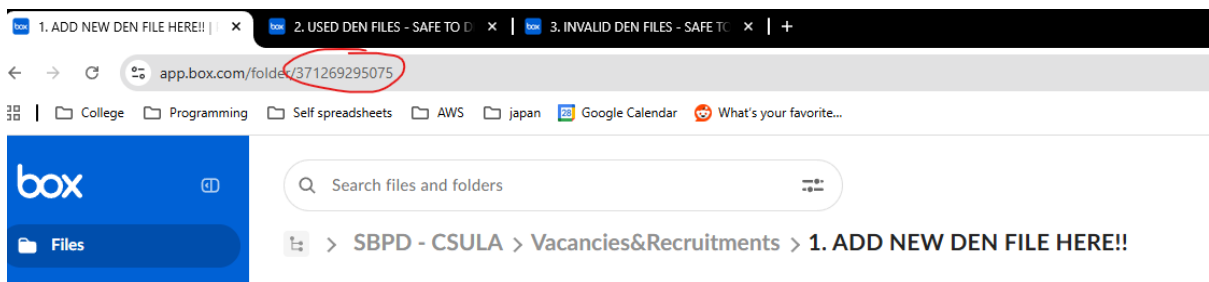


Figure 34: Vacancies&Recruitments Project - Box Upload Folder ID

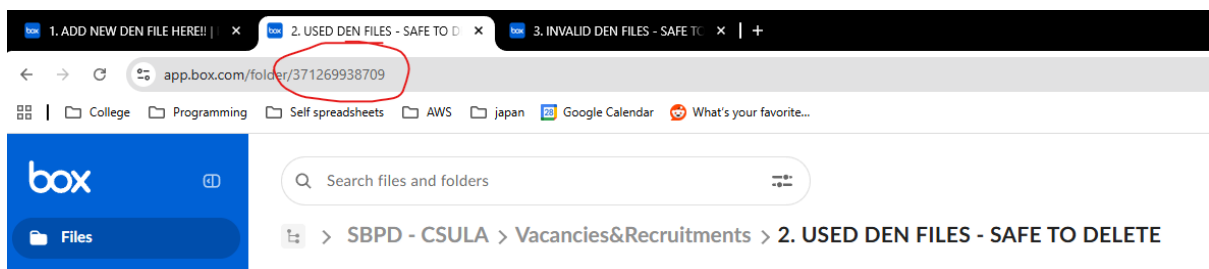


Figure 35: Vacancies&Recruitments Project - Box Used Folder ID

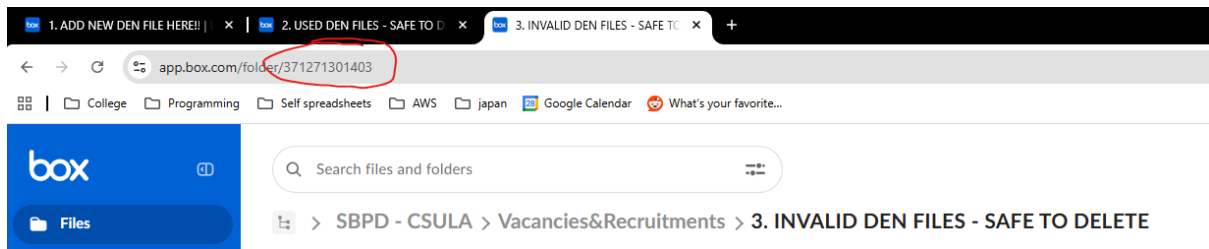


Figure 36: Vacancies&Recruitments Project - Box Invalid Folder ID

3. Overall, this is what the Vacancies&Recruitments class should look like.

```

77 #####
78 # Vacancies Project
79 #####
80 class Vacancies:
81     class Smartsheet:
82         VACANCIES_TABLE_ID = 4963448636002180
83
84     class Box:
85         DEN_UPLOAD_FOLDER_ID = 371269295075
86         USED_DEN_FILES_FOLDER_ID = 371269938709
87         INVALID_DEN_FILES_FOLDER_ID = 371271301403
88 
```

Figure 37: Vacancies&Recruitments Class Config

11.5.5. Filling in Personnel Matters Configuration class

1. List Personnel Matters Column IDs.

> List Personnel Matters Column IDs

```
? Smartsheet IDs CLI List Personnel Matters Column IDs
Table: [Personnel Matters] Tracker - 1234840859922308
Column Name | Column ID
-----+-----
MatterID | 3548716836016004
Matter IDs | 8052316463386500
Opened Date | 733967068909444
Status | 5237566696279940
Complaint(s) | 2985766882594692
Respondent(s) | 7489366509965188
What Happened | 1859866975752068
Issue | 6363466603122564
Policy | 4111666789437316
Intake Submitted Via | 8615266416807812
Supervisor | 4956091719569284
Next Steps | 2704291905884036
Next Step Due Date | 7207891533254532
Memo Notes | 1578391999041412
Closed Date | 6081991626411908
Outcome | 3830191812726660
Modified By | 8333791440097156
Last Modified Date | 1015442045620100
Fiscal Year | 5519041672990596
Update Required | 3267241859305348
Days Open | 7770841486675844
Missing Close Info | 2141341952462724
Box Sync Status | 4580764537884548
Box Folder Link | 2328964724199300
Box Uploaded At | 6832564351569796
Box Sync Notes | 1203064817356676
Box Synced Attachment IDs | 5706664444727172
Do you have documents to attach? | 6644941579833220
Documents Link | 452492092198788
```

Figure 38: smartsheet_ids_cli.py Listing Personnel Matters column IDs

2. Navigate to Box.com's Personnel Matters folder and attach that folder.

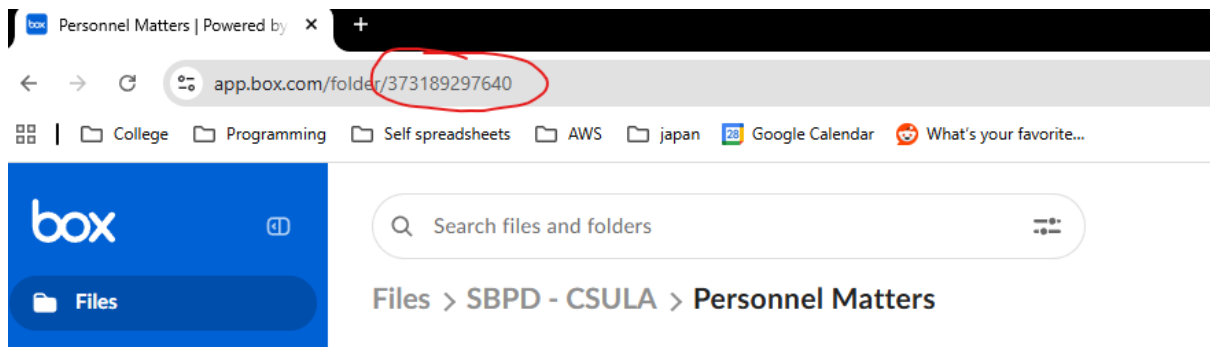


Figure 39: Personnel Matters Project - Box Personnel Matters Folder ID

- Overall, this is what the Personnel Matters class should look like.

```

90      #####
91      # Personnel Matters Project
92      #####
93      class PersonnelMatters:
94          class Smartsheet:
95              PERSONNEL_MATTERS_TABLE_ID = 1234840859922308
96              MATTER_COLUMN_IDS = 8052316463386500
97              RESPONDENT_COLUMN_ID = 7489366509965188
98              BOX_SYNC_STATUS_COLUMN_ID = 4580764537884548
99              DO_YOU_HAVE_ANY_DOCUMENTS_COLUMN_ID = 6644941579833220
100
101          class Box:
102              PERSONNEL_MATTERS_BOX_ROOT_FOLDER_ID = 373189297640
103

```

Figure 40: Personnel Matters Class Config

11.6. Deploying AWS Stacks using AWS SAM

Deploying the AWS Stacks allows the python scripts to be hosted on AWS Lambda, which are Amazon's serverless compute instances. Doing this allows Smartsheet and Box.com to send requests to our AWS lambdas and let them know when a new event has occurred, causing the scripts to execute without manually intervention.

- Repackage the layers into a zip file. Anytime any changes are made to shared_config/ folder, the parent python folder needs to be re-zipped. Then the following commands must be run.

```
sam build
```

```
sam deploy
```

- After saying "yes" to their stack changes, the cloud formation stacks should create or update the stacks in AWS. Which will either create or update the API Gateway, IAM roles, AWS Lambdas, and Secrets Manager services.

3. Next, in order to give the lambdas credentials to run the script, you need to upload the .env credentials into AWS SecretsManager.

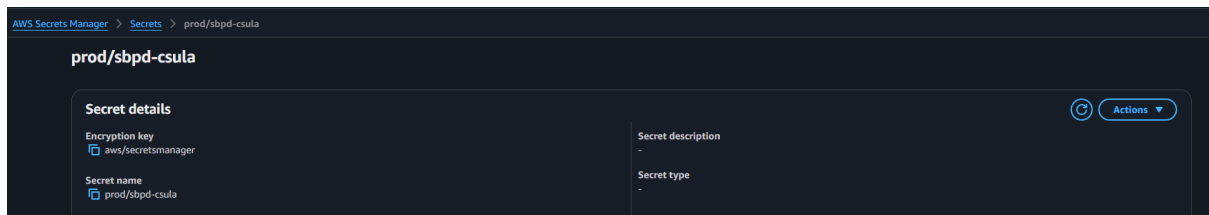


Figure 41: AWS Secrets Manager

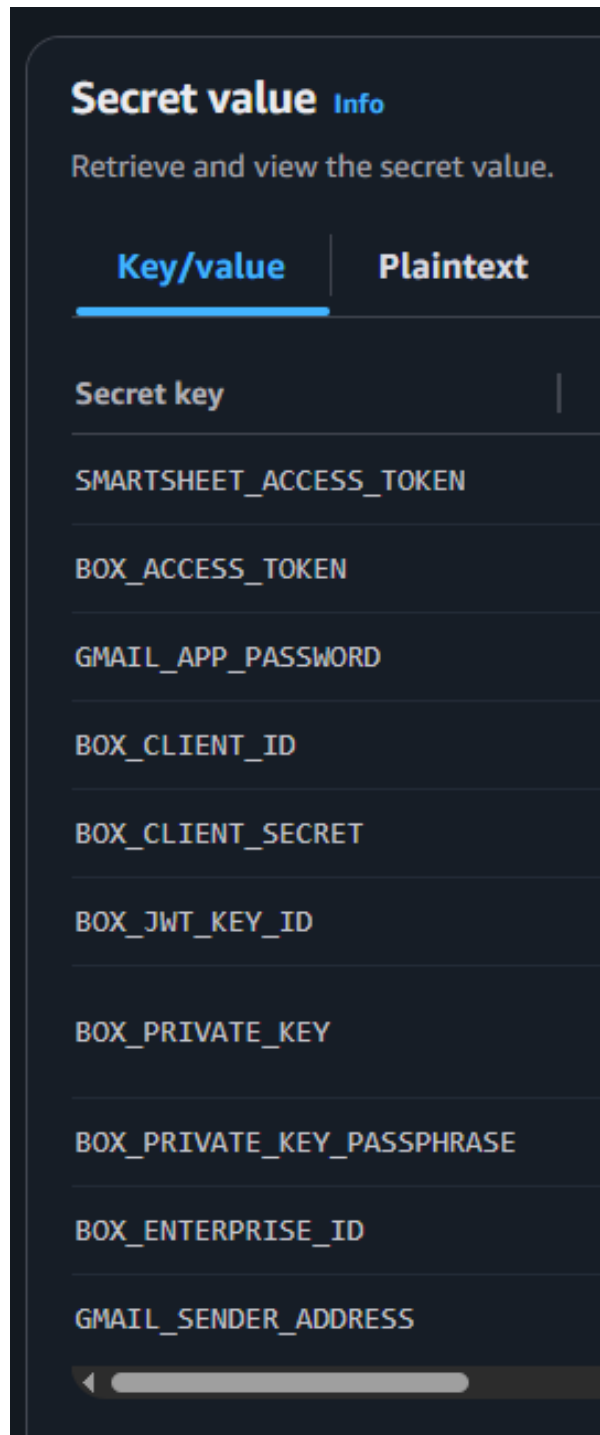


Figure 42: AWS Secrets Manager

4. Before we turn on the webhooks that allow Smartsheet and Box.com to send requests to our lambdas, we first need to add the endpoints of our lambdas to our .env file. And we can find the endpoints by navigating to AWS API Gateway.

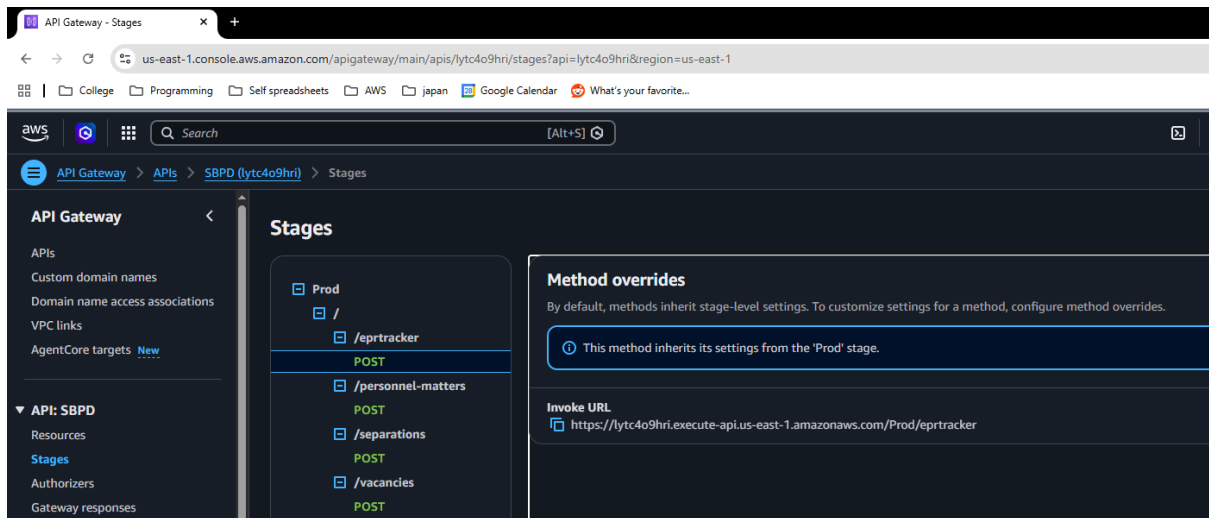


Figure 43: AWS API Gateway endpoints

```

5 # Webhook Settings
6 EPR_TRACKER_API_GATEWAY_ADDRESS="https://lytc4o9hri.execute-api.us-east-1.amazonaws.com/Prod/epctracker"
7 PERSONNEL_MATTERS_API_GATEWAY_ADDRESS="https://lytc4o9hri.execute-api.us-east-1.amazonaws.com/Prod/personnel-matters"
8 SEPARATIONS_API_GATEWAY_ADDRESS="https://lytc4o9hri.execute-api.us-east-1.amazonaws.com/Prod/separations"
9 VACANCIES_API_GATEWAY_ADDRESS="https://lytc4o9hri.execute-api.us-east-1.amazonaws.com/Prod/vacancies"
10

```

Figure 44: .env AWS API Gateway endpoints

5. Finally, we can turn on all the webhook endpoints by running the following commands.

```
python webhook_cli.py
```

```
> Create Webhook
```

```
> SMARTSHEET) EPR Tracker
```

```
> SMARTSHEET) Personnel Matters
```

```
> SMARTSHEET) Separations Tracker
```

```
> BOX) Vacancies & Recruitment Tracker
```

12. Glossary

- **EPR (Employee Performance Review):** A structured evaluation process for employee performance and development.
- **HR (Human Resources):** The department responsible for employee records, reviews, and workflow management.
- **Smartsheet:** Cloud-based platform used to manage employee data, workflow automation, and EPR tracking.
- **Box:** Cloud storage system used to archive finalized and signed EPR documents.
- **Power BI:** Data analytics tool used to visualize EPR metrics, progress, and workforce summaries.
- **Power Automate:** Automation service used to link Smartsheet, DocuSign, Box, and Power BI for workflow synchronization.
- **Dashboard:** A Power BI visual interface displaying HR analytics and EPR tracking metrics.
- **OAuth 2.0:** Authentication method used to ensure secure access between connected services.
- **KPI (Key Performance Indicator):** Quantifiable metric used to measure progress toward HR goals.
- **SRS (Software Requirements Specification):** The document that defines all functional and non-functional system requirements.
- **SDS (Software Design Specification):** This document; describes how the system is structured, integrated, and implemented.

13. References

- Software Requirements Specification – Intelligent HR Analytics & EPR Management Dashboard
- Smartsheet Product Documentation: <https://help.smartsheet.com/>
- Microsoft Power BI Documentation: <https://learn.microsoft.com/power-bi/>
- Box API and Security Overview: <https://developer.box.com/>
- Santa Barbara County Public Defender’s Office – HR Process Guidelines (Internal Reference)
- Ascent (CSULA Senior Design Project Portal) – Team Sun-1 Documentation (2025): <https://ascent.cysun.org/project/project/view/251>